# ROS Spin Professor Handoff

## Corrected Doxorubicin Semiquinone Oxygen Spin Chemistry Model

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Professor-facing final computational report

All quantitative figures and tables in this document were generated directly from the ROS_Spin computational workflow or from values stored in its machine-readable parameter and provenance files. No figures were reproduced from external publications. Figure 1 is an author-created conceptual workflow and contains no independent quantitative data.

**NON-PREDICTIVE DIMENSIONLESS SENSITIVITY ANALYSIS**

# Purpose of This Handoff

This report hands off the completed corrected ROS_Spin implementation, its generated scientific results, numerical validation, evidence record, and reproduction package. It is written to support technical review and manuscript revision without requiring the reader to reconstruct the repository history.

All quantitative figures and tables in this document were generated directly from the ROS_Spin computational workflow or from values stored in its machine-readable parameter and provenance files. No figures were reproduced from external publications. Figure 1 is an author-created conceptual workflow and contains no independent quantitative data.

# Executive Summary

Across the five baseline initial-state benchmarks, the computed primary-superoxide yield per encounter spans 0.140445 to 0.445496. Across the selected sensitivity grids it spans 0.000594 884 to 1; these extrema describe only the executed grid. The simulations demonstrate how primary superoxide yield would depend on the competition among spin-selective electron transfer, doublet-quartet mixing, relaxation, and encounter escape under specified dimensionless scenarios. They do not establish that the assumed spin preparation, mixing magnitude, or state-selective rates occur in the doxorubicin semiquinone-oxygen system.

Computed structural results include projector algebra, six-state dimensions, valid density matrices, probability accounting, solver agreement, and coherent-embedding consistency. Conditional sensitivity results cover mixing, relaxation, escape, initial state, and kQ/kD. Literature-derived calculations remain confined to condition-specific bulk rates, downstream aqueous kinetics, and provenance-qualified experimental context.

The worst absolute discrepancy between the reference matrix exponential and the separately constructed adaptive Dormand-Prince solver is 1.044e-12. The maximum encounter probability-balance error is 1.318e-13, and the maximum downstream radical-equivalent balance error is 1.694e-21 M. These checks validate numerical consistency rather than physical parameterization.

Circuit status for the recorded run is qiskit_executed. The three-qubit calculation embeds the six physical states within an eight-state space, checks the two unused states for leakage, and compares statevectors and doublet/quartet observables. It validates coherent encoding consistency only; it does not demonstrate quantum advantage or independently validate the chemistry.

# Connection to the Original ROS Spin Project

ROS_Spin originally asked whether electron-spin dynamics could influence reactive oxygen species formation during anthracycline redox cycling while retaining a quantum-circuit component for computational comparison. The completed workflow preserves that objective but replaces the original two-qubit chemical interpretation with the correct spin space for doxorubicin semiquinone and molecular oxygen.

The prior implementation used singlet/triplet and Bell-state ideas for a system whose oxygen partner has spin one. It also risked treating computational-basis parity and repeated circuit gates as chemical dynamics, carrying semiconductor relaxation data into an unrelated molecular encounter, and mapping spin outcomes directly to ROS products. Those choices could not support the proposed chemistry.

The corrected workflow keeps the biological motivation at the level supported by current evidence: anthracycline semiquinone chemistry can transfer an electron to oxygen, and downstream superoxide chemistry can form H2O2. The present encounter model does not bridge its illustrative per-encounter yields to organelle, cell, animal, or patient exposure without missing formation, association, residence, competing-sink, transport, and compartment parameters.

# Table 1 Original versus Corrected Implementation

Table 1 connects the corrected implementation to the original project and identifies why each change matters scientifically.

| Model element | Original implementation | Corrected implementation | Consequence |
|---|---|---|---|
| Physical species | Generic two-spin or semiconductor shuttling model | Doxorubicin/adriamycin semiquinone with S=1/2 and ground-state O2 with S=1 | The chemistry is tied to the stated redox pair. |
| Spin dimensions | Two qubits; dimension 4 | Spin-1/2 x spin-1 electronic space; dimension 6 | All physical electronic states are retained. |
| Manifold classification | Singlet/triplet or basis-parity labels | Doublet (dimension 2) and quartet (dimension 4) projectors | Projectors follow angular-momentum addition. |
| Initial states | Bell states or computational-basis preparations | I6/6, PD/2, PQ/4, and bounded pD mixtures | All cases are computational benchmarks; no chemical preparation is asserted. |
| Hamiltonian | Repeated CZ or identity gates used as chemical dynamics | Zeeman, exchange, dipolar, O2 ZFS, and local-field terms in the six-state space | Unavailable interactions remain disabled or sensitivity-only. |
| Relaxation | Semiconductor dephasing parameters | Local isotropic Lindblad sensitivity model with exact-system rates unavailable | No semiconductor relaxation parameter enters active chemistry. |
| Electron transfer | Mapped from state counts or parity | Integrated kD Tr(PD rho) and kQ Tr(PQ rho) loss fluxes | Reaction yield is a time-integrated kinetic quantity. |
| Encounter escape | Absent or not separated from circuit depth | Independent first-order escape channel kescape Tr(rho) | Reaction, escape, and unresolved survival close the probability balance. |
| Primary superoxide | Assigned from singlet/triplet or basis outcomes | YD + YQ from integrated electron-transfer flux | One primary radical equivalent is assigned per reacted encounter. |
| Hydrogen peroxide | Mapped directly from a spin population | Separate HO2/O2-minus speciation and dismutation with two radicals per H2O2 | Spin populations are not treated as H2O2. |
| Rate units | Bulk and encounter rates could be conflated | Bulk M^-1 s^-1 constants remain separate from encounter s^-1 rates | No unsupported dimensional conversion is made. |
| Quantum-circuit role | Circuit outcomes interpreted as chemistry | Three-qubit coherent embedding consistency check | The circuit does not validate open-system chemistry or quantum advantage. |
| Interpretation | Risk of direct biological or cardiotoxicity extrapolation | Conditional dimensionless sensitivity analysis with evidence-gated inputs | Cellular ROS and clinical outcomes remain outside scope. |

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table01_original_versus_corrected_model.csv

# Corrected Model and Equations

Doxorubicin or adriamycin semiquinone has electron spin S=1/2, whereas ground-state O2 has S=1. Their product space therefore contains 2 x 3 = 6 electronic states. This change determines the correct state dimension and invalidates a two-qubit singlet/triplet description of the encounter.

Angular-momentum addition splits the six-state space into a doublet subspace of dimension two and a quartet subspace of dimension four. Exact projectors define the manifold observables. I6/6, PD/2, PQ/4, and bounded pD mixtures are computational benchmarks, not evidence of chemical state preparation.

$$P_D = \frac{0.5I - \mathbf{s}\cdot\mathbf{S}}{1.5}, \qquad P_Q = \frac{\mathbf{s}\cdot\mathbf{S} + I}{1.5}$$

The active model propagates a six-by-six density matrix with a constant-generator matrix exponential. Zeeman, exchange, dipolar, oxygen zero-field-splitting, and local-field structures are implemented, but unavailable exact-system terms remain disabled or are activated only as named sensitivity coordinates. Local isotropic Lindblad terms supply an explicitly phenomenological relaxation sensitivity model.

$$\frac{H}{\hbar} = \beta_e \mathbf{B}\cdot(g_{\mathrm{SQ}}\mathbf{s} + g_{\mathrm{O_2}}\mathbf{S}) + J\mathbf{s}\cdot\mathbf{S} + \mathbf{s}\cdot\mathbf{D}_{\mathrm{dip}}\cdot\mathbf{S} + \mathbf{S}\cdot\mathbf{D}_{\mathrm{O_2}}\cdot\mathbf{S} + \mathbf{\Omega}_{\mathrm{local}}\cdot\mathbf{s}$$

Electron transfer is represented by separate first-order loss operators in the doublet and quartet manifolds, while encounter escape removes population independently. The model integrates these fluxes through time and retains unresolved survival so that reaction, escape, and survival close to unity.

$$\frac{d\rho}{dt} = -i[H/\hbar, \rho] - \frac{1}{2}\{k_D P_D + k_Q P_Q + k_{\mathrm{escape}} I, \rho\} + \mathcal{L}_{\mathrm{relax}}(\rho)$$

Primary superoxide is calculated only as the sum of integrated doublet and quartet electron-transfer yields. One reacted encounter contributes one primary superoxide-family radical equivalent. A final spin population is never re-labeled as chemical product.

$$Y_D(t) = \int_0^t k_D \mathrm{Tr}[P_D \rho(\tau)]\,d\tau, \quad Y_Q(t) = \int_0^t k_Q \mathrm{Tr}[P_Q \rho(\tau)]\,d\tau, \quad Y_{\mathrm{primary}} = Y_D + Y_Q$$

$$Y_D + Y_Q + Y_{\mathrm{escape}} + \mathrm{Tr}(\rho) = 1$$

A distinct fixed-pH model partitions the radical pool between HO2 and O2-minus, applies spontaneous and SOD-mediated dismutation, consumes two radical equivalents per H2O2, and tracks an optional H2O2 loss channel. This stage does not feed an encounter spin population directly into H2O2.

$$2\ \text{radical equivalents} \longrightarrow 1\ \mathrm{H_2O_2}$$

# Figure 1 Conceptual Model Overview

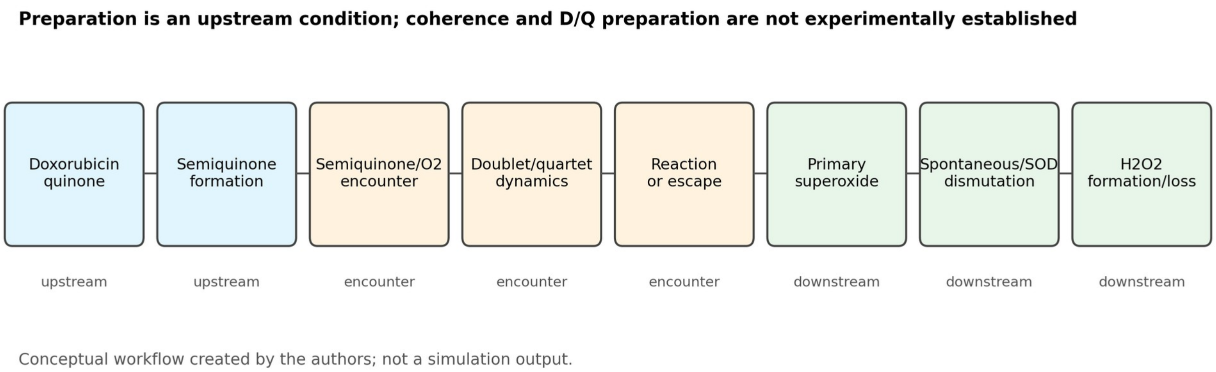



**Figure 1 Corrected Model Overview.** Corrected model overview. Doxorubicin reduction and semiquinone formation are classical upstream stages. A semiquinone-triplet-O2 encounter enters the six-state doublet/quartet quantum spin-evolution stage, followed by competing spin-selective electron transfer or escape. Integrated reaction flux forms primary superoxide; fixed-pH HO2/O2-minus speciation and dismutation subsequently form H2O2, which may be lost through a separate sink. Conceptual workflow created by the authors; not a simulation output. The diagram is non-quantitative and does not assert coherent preparation or a measured encounter.

**Source data** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure01_model_overview.csv **Caption** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/captions/figure01_model_overview_caption.txt

# Computational Methods and Script Mapping

The active workflow separates model equations, command-line policy, sensitivity design, plotting, evidence, and tests so that each generated claim can be traced to code and source data.

**spin_chemistry.py** Defines spin operators and projectors, six-state Hamiltonians, reference and independent propagators, staged ROS chemistry, evidence gates, and coherent embedding checks.

**ROS.py** Provides the guarded command line for a single evidence-backed bulk calculation or an explicitly opted-in encounter sensitivity case.

**sensitivity_analysis.py** Defines normalized one- and two-dimensional sweeps, benchmark initial states, probability outputs, and the non-predictive labeling policy.

**paper_analysis.py** Runs the deterministic paper workflow, writes Figure 1-11 source data, renders all figures, generates Tables 1-9, and records run metadata.

**plotting.py** Renders each figure from its saved source CSV using one visual style and the required sensitivity warning.

**configs/doxorubicin_parameters.json** Stores active, disabled, sensitivity-only, downstream, bulk-rate, and experimental-evidence records.

**configs/parameter_provenance.csv** Provides the machine-readable source ledger, conditions, uncertainty, suitability, and exact code paths.

**tests** Checks spin algebra, dynamics, evidence policy, independent validation, downstream balances, circuit behavior, legacy boundaries, and the paper and handoff pipelines.

The JSON configuration and provenance CSV distinguish measured or literature-derived bulk quantities from missing encounter quantities. Bulk constants in M^-1 s^-1 remain separate from kD, kQ, relaxation, and escape in s^-1. Missing geometry, tensors, relaxation times, state preparation, and state-resolved rates are not assigned invented values.

All encounter results are conditional, dimensionless sensitivity results. Mixing, relaxation, escape, initial doublet fraction, and kQ/kD are varied over declared computational coordinates. The equality kQ/kD=1 is the spin-independent null; the displayed axes are not measured ranges, probability distributions, priors, or confidence intervals.

# Figure 2 Benchmark Trajectories

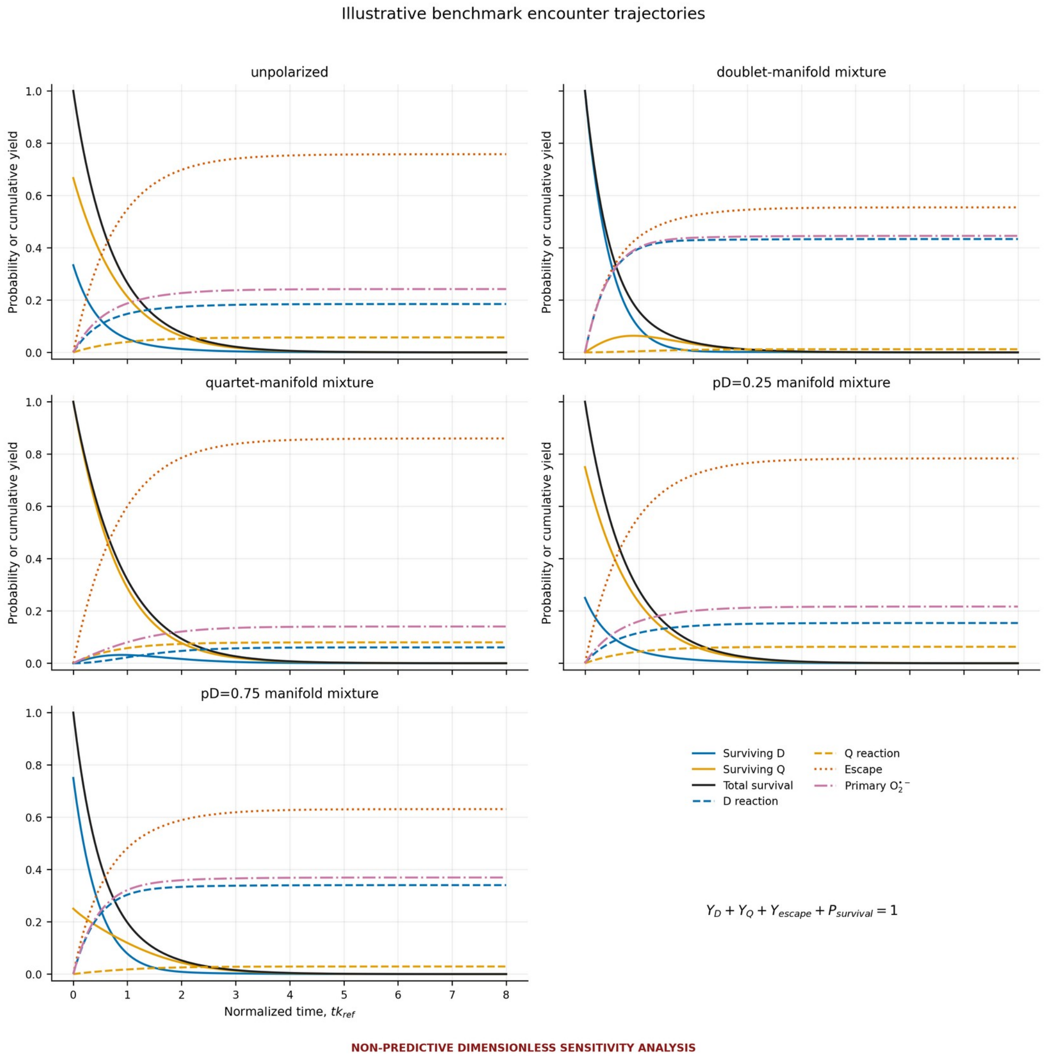



**Figure 2 Benchmark Encounter Trajectories.** Benchmark encounter trajectories. Surviving doublet and quartet populations, total survival, cumulative doublet and quartet reaction yields, primary-superoxide yield, and escape yield are shown against normalized time for five benchmark mixtures. At every sampled time, YD + YQ + Yescape + Psurvival = 1 within numerical tolerance. The settings kD/kref=1, kQ/kD=0.1, kescape/kref=1, omega_local/kref=1, and gammaR/kref=gammaO/kref=0.1 are illustrative sensitivity coordinates, not measured parameters.

**Source data** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure02_benchmark_trajectories.csv **Caption** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/captions/figure02_benchmark_trajectories_caption.txt

# Figure 3 Initial State Comparison

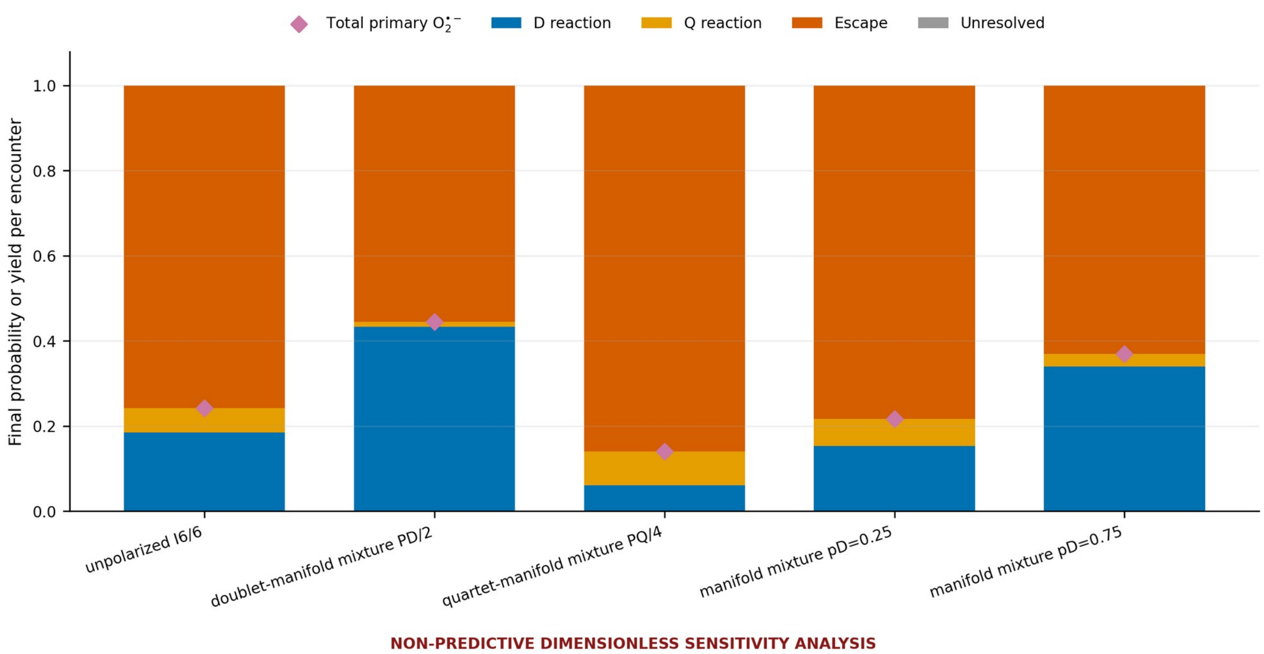



**Figure 3 Initial State Comparison.** Initial-state comparison. Final reaction, escape, and unresolved survival outcomes are compared for the unpolarized encounter, doublet-manifold benchmark, quartet-manifold benchmark, and pD=0.25 and pD=0.75 mixtures. The doublet and quartet cases are computational limiting benchmarks rather than demonstrated chemically prepared states. Primary superoxide is the sum of integrated D and Q reaction yields, not a final spin population.

**Source data** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure03_initial_state_comparison.csv **Caption** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/captions/figure03_initial_state_comparison_caption.txt

# Figures 4 and 5 Timescale Competition

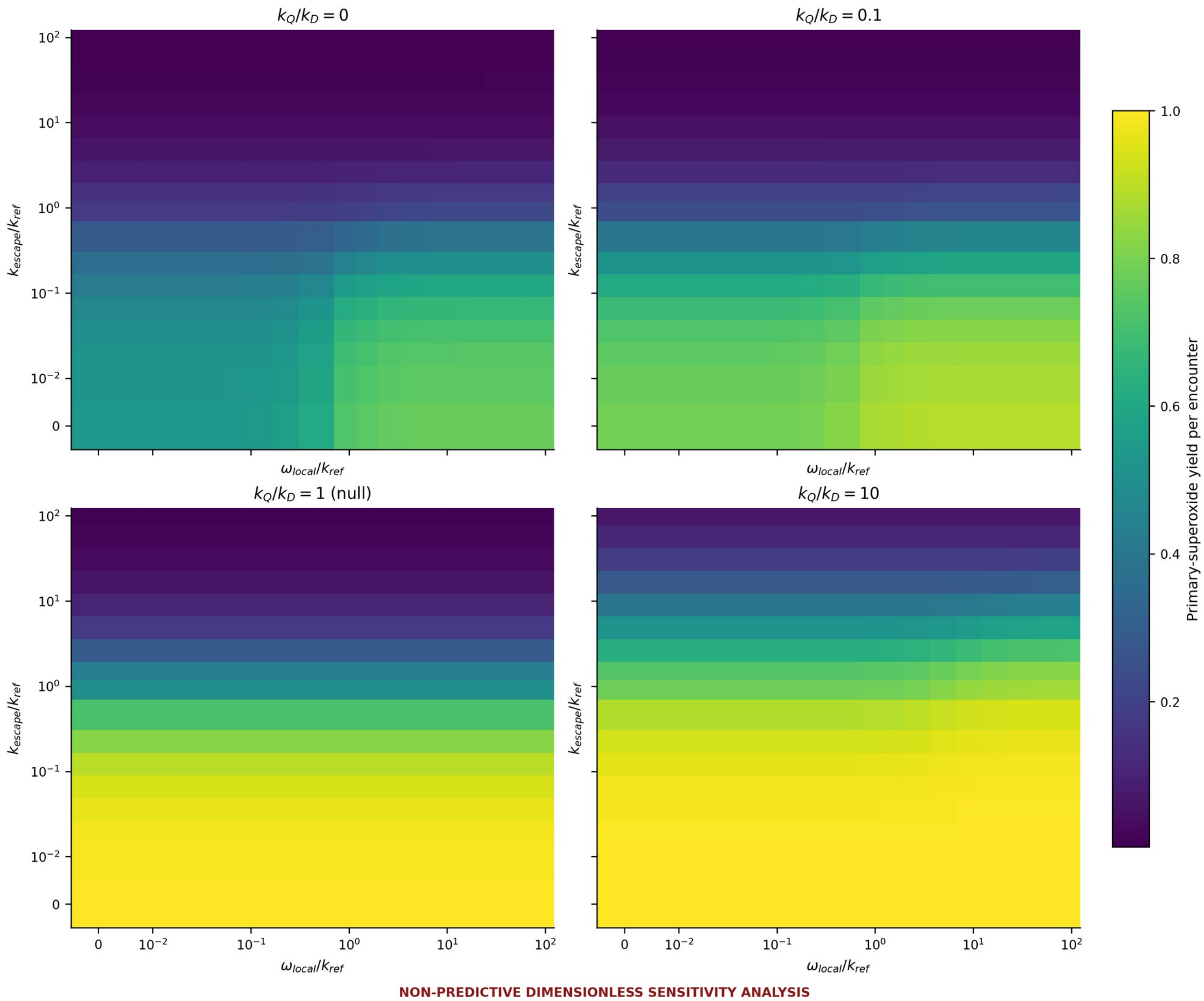



**Figure 4 Mixing versus Escape Heatmaps.** Mixing-versus-escape heatmaps. Primary-superoxide yield per encounter is calculated over normalized local D/Q mixing and escape rates for kQ/kD=0, 0.1, 1, and 10. The kQ/kD=1 panel is the spin-independent null condition. Zero and logarithmic nonzero coordinates are illustrative; no displayed range is experimentally established.

**Source data** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure04_mixing_escape_heatmaps.csv **Caption** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/captions/figure04_mixing_escape_heatmaps_caption.txt

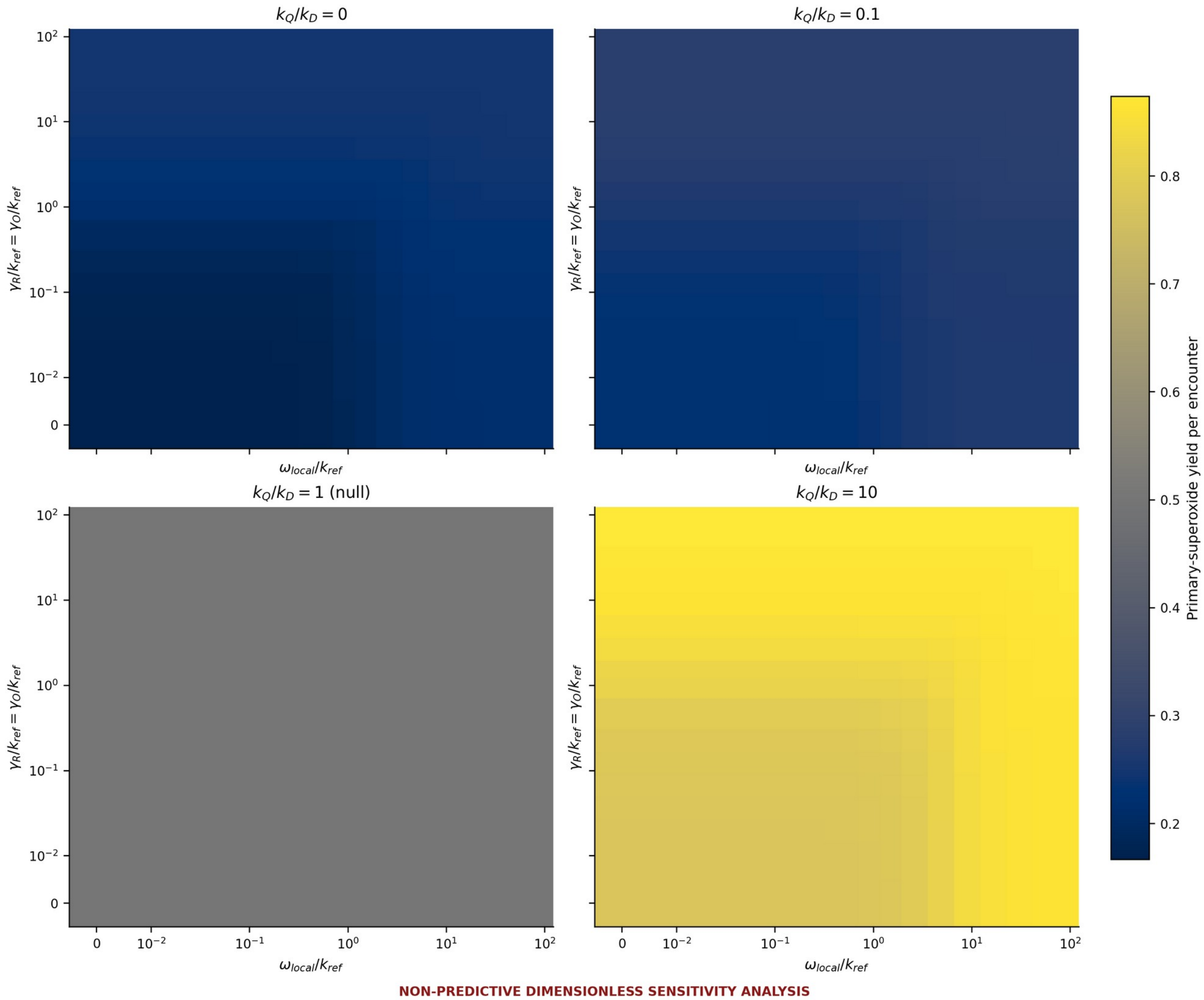



**Figure 5 Mixing versus Relaxation Heatmaps.** Mixing-versus-relaxation heatmaps. Primary-superoxide yield is calculated for an unpolarized benchmark over normalized local mixing and equal radical/O2 relaxation rates. The fixed settings are kD/kref=1, kescape/kref=1, and duration=8/kref; panels show kQ/kD=0, 0.1, 1, and 10, with equality marked as the spin-independent null. All coordinates are illustrative and non-predictive.

**Source data** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure05_mixing_relaxation_heatmaps.csv **Caption** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/captions/figure05_mixing_relaxation_heatmaps_caption.txt

# Figure 6 Reaction Selectivity Sensitivity

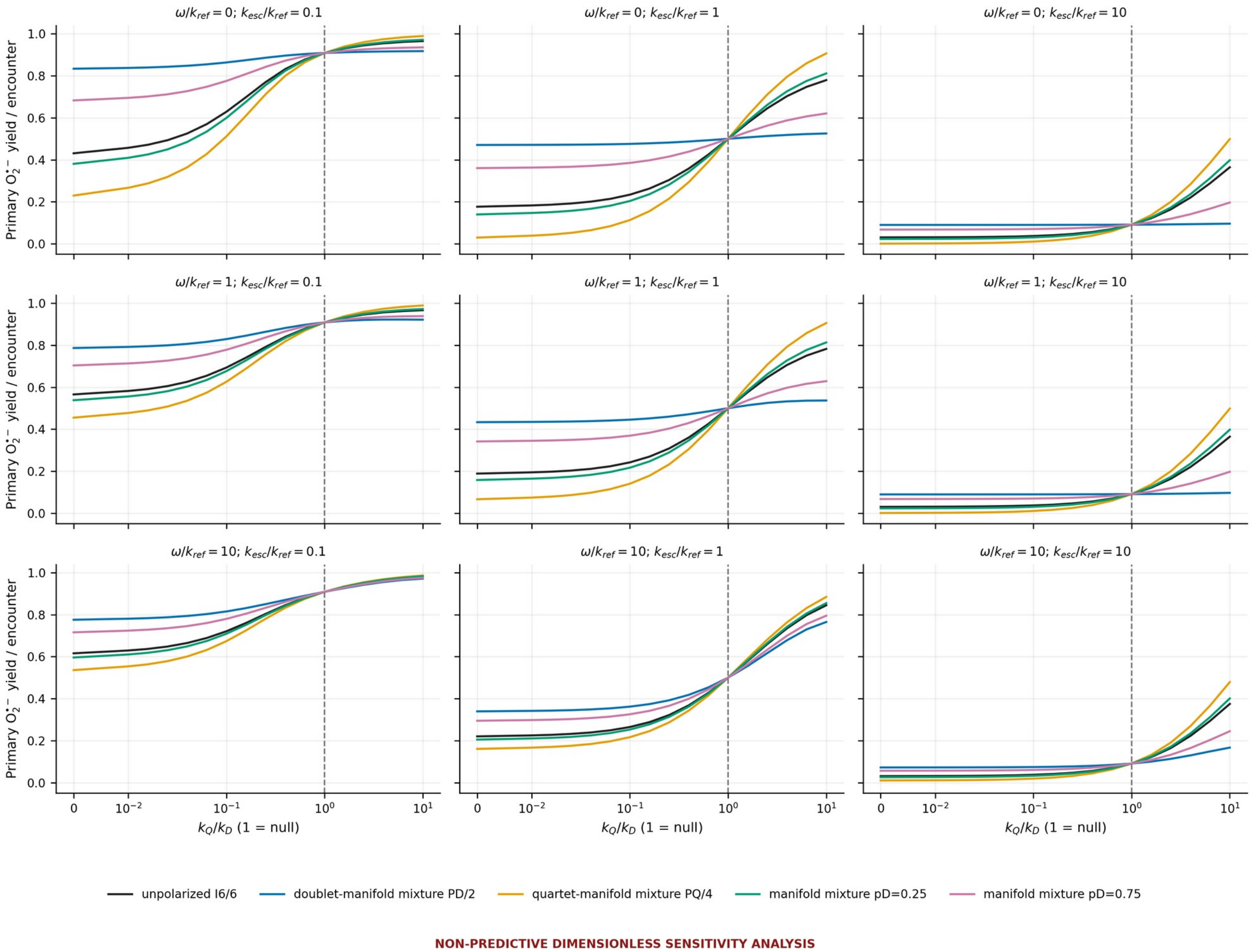



**Figure 6 Reaction Selectivity Sensitivity.** Reaction-selectivity sensitivity. Primary-superoxide yield is plotted against kQ/kD for unpolarized, doublet, quartet, pD=0.25, and pD=0.75 benchmarks across three mixing and escape settings. The vertical line marks kQ/kD=1 as the spin-independent null. The kQ/kD axis is a sensitivity coordinate, not a measured or chemically defensible probability distribution or physical range.

**Source data** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure06_reaction_selectivity.csv **Caption** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/captions/figure06_reaction_selectivity_caption.txt

# Figure 7 Controls

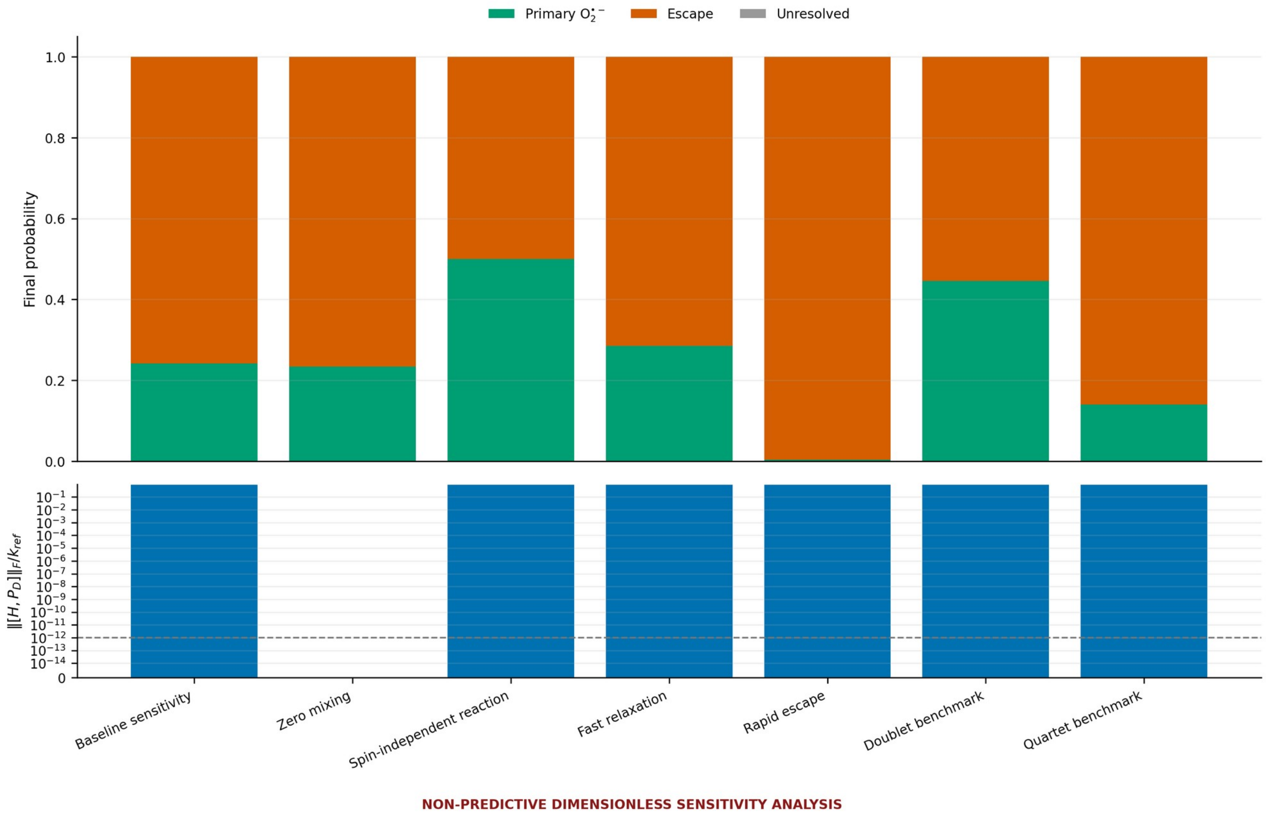



**Figure 7 Controls and Limiting Cases.** Controls and limiting cases. Baseline sensitivity, zero mixing, kQ=kD spin-independent reaction, fast relaxation, rapid escape, doublet benchmark, and quartet benchmark cases show primary reaction, escape, and unresolved survival. Stacked outcomes make probability accounting visible. All rates and initial-state restrictions are computational controls, not established encounter parameters or preparations.

**Source data** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure07_controls.csv **Caption** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/captions/figure07_controls_caption.txt

# Figure 8 Independent Solver Validation

The worst absolute discrepancy between the reference matrix exponential and the separately constructed adaptive Dormand-Prince solver is 1.044e-12. The maximum encounter probability-balance error is 1.318e-13, and the maximum downstream radical-equivalent balance error is 1.694e-21 M. These checks validate numerical consistency rather than physical parameterization.

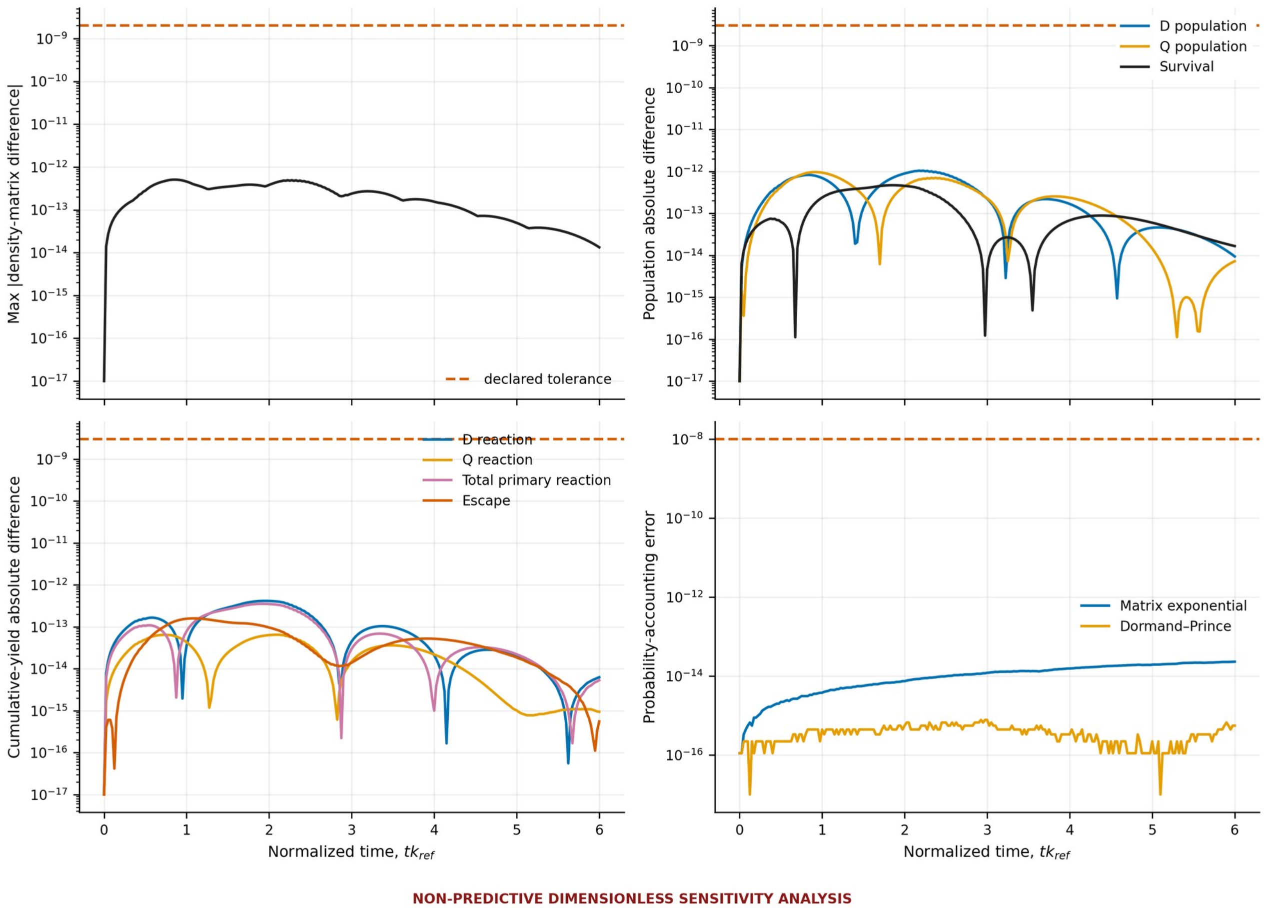



**Figure 8 Independent Numerical Validation.** Independent numerical validation. A separately constructed adaptive Dormand-Prince 5(4) solver is compared with the constant-generator matrix-exponential reference for full density matrices, doublet and quartet populations, survival, reaction yields, escape yield, and probability balance. Dashed lines show declared tolerances. Agreement validates numerical implementation only, not physical encounter inputs.

**Source data** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure08_solver_validation.csv **Caption** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/captions/figure08_solver_validation_caption.txt

# Figure 9 Downstream ROS Kinetics

A distinct fixed-pH model partitions the radical pool between HO2 and O2-minus, applies spontaneous and SOD-mediated dismutation, consumes two radical equivalents per H2O2, and tracks an optional H2O2 loss channel. This stage does not feed an encounter spin population directly into H2O2.

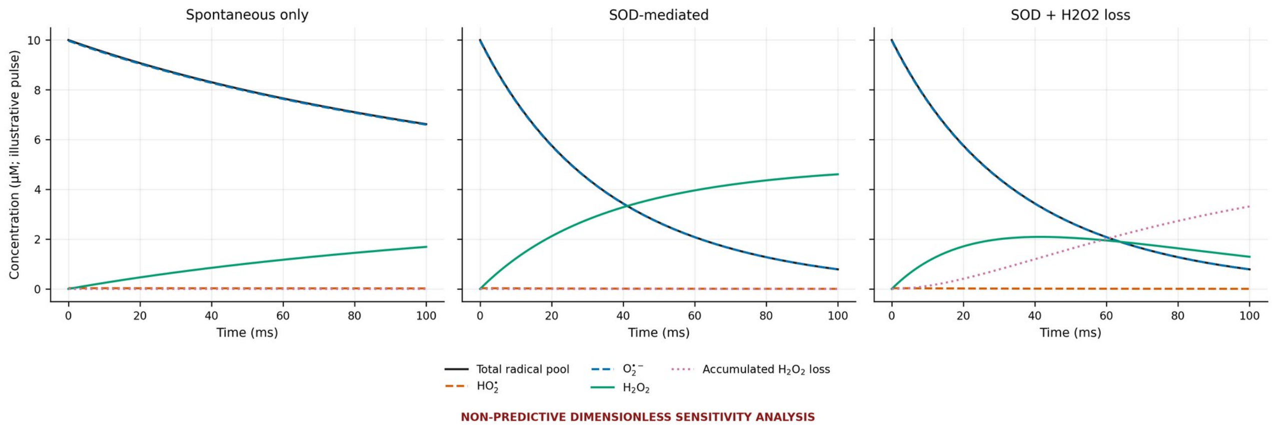
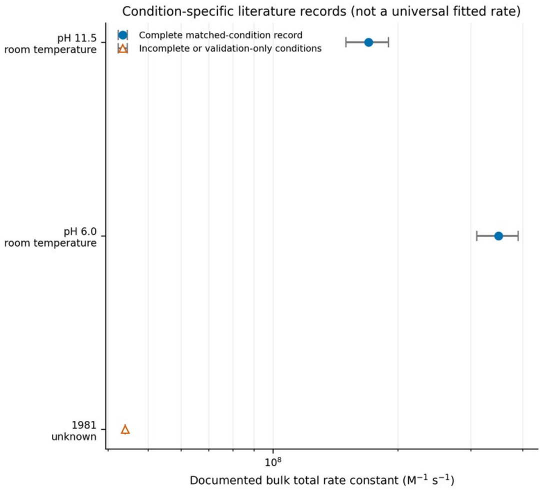



**Figure 9  Downstream ROS Kinetics.** Downstream ROS kinetics. Total superoxide-family radical pool, HO2, O2-minus, accumulated H2O2, and accumulated H2O2 loss are shown for spontaneous dismutation, SOD-mediated dismutation, and SOD plus H2O2 loss. The calculation starts from an illustrative 10 micromolar single radical pulse at fixed pH 7.4, assumes rapid acid-base equilibrium and constant rates/SOD, and consumes two radical equivalents per H2O2. No spin population is mapped directly to H2O2, and the trajectories are not cellular predictions.

**Source data** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure09_downstream_kinetics.csv **Caption** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/captions/figure09_downstream_kinetics_caption.txt

# Figure 10 Primary Source Bulk Rate Context

The plotted literature values remain condition-specific bulk kinetic records. Their units and incomplete conditions prevent their use as encounter-level state-resolved reaction rates.




Records remain conditionally distinct

| Record | Conditions/environment | Species/protonation | Source/allowed use |
|---|---|---|---|
| 1981 | pH unknown; unknown; aqueous pulse radiolysis | adriamycin semiquinone plus ground-state triplet O2; protonation: unknown | https://doi.org/10.1016/0003-9861(81)90263-0; allowed: literature_arithmetic;sensitivity_only |
| pH6 | pH 6.0; room temperature; aqueous phosphate/formate; O2 or O2/N2 purged | adriamycin semiquinone plus neutral ground-state triplet O2; protonation: biological-pH internally protonated/zwitterionic semiquinone; O2 neutral | https://doi.org/10.1038/bjc.1985.74; allowed: literature_arithmetic;conditi on_matched_prediction;sensitivity_only |
| pH11_5 | pH 11.5; room temperature; aqueous borate/OH-; O2 or O2/N2 purged | deprotonated adriamycin semiquinone plus neutral ground-state triplet O2; protonation: deprotonated semiquinone; O2 neutral | https://doi.org/10.1038/bjc.1985.74; allowed: literature_arithmetic;conditi on_matched_prediction;sensitivity_only |

**Figure 10  Literature Bulk Rate Comparison.** Literature bulk-rate comparison. Actual numerical values stored in the configuration JSON and matched provenance CSV are plotted with reported uncertainty, pH, temperature, environment, species/protonation, method, and source identifiers. Missing conditions remain visibly marked and records are not combined into a universal rate. These M^-1 s^-1 bulk total constants are not encounter-level kD or kQ in s^-1.

**Source data** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure10_bulk_rate_comparison.csv **Caption** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/captions/figure10_bulk_rate_comparison_caption.txt

# Figure 11 Circuit Embedding Validation

Circuit status for the recorded run is qiskit_executed. The three-qubit calculation embeds the six physical states within an eight-state space, checks the two unused states for leakage, and compares statevectors and doublet/quartet observables. It validates coherent encoding consistency only; it does not demonstrate quantum advantage or independently validate the chemistry.

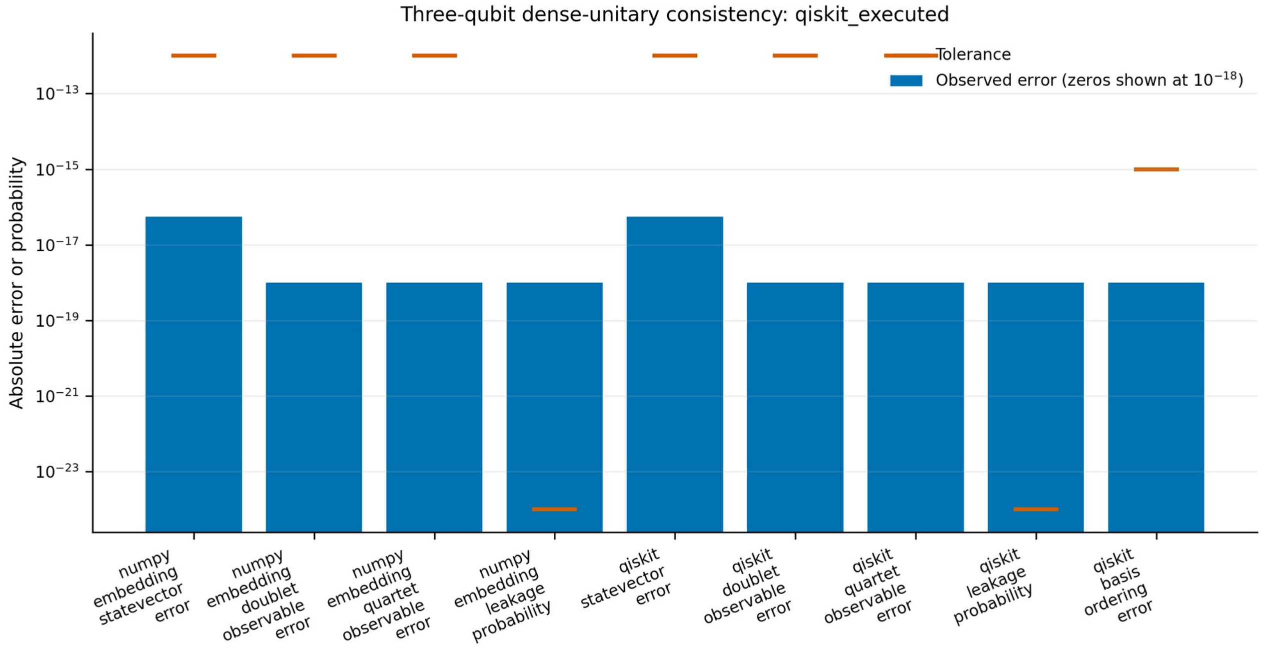



**Figure 11  Three Qubit Coherent Embedding Validation.** Three-qubit coherent-embedding validation. The six-state coherent reference unitary is compared with its eight-state three-qubit embedding and, when requested and available, actual Qiskit statevector execution. Statevector, doublet and quartet observable, basis-ordering, and unused-state leakage errors are compared with declared tolerances. This validates coherent simulator and encoding consistency only; it does not validate reaction, relaxation, escape, downstream chemistry, hardware performance, or quantum advantage.

**Source data** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure11_circuit_validation.csv **Caption** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/captions/figure11_circuit_validation_caption.txt

# Baseline and Control Tables

Tables 4-7 report the generated baseline, controls, independent-solver errors, and coherent-embedding metrics. All numeric cells are populated from their CSV sources.

## Table 4 Baseline Results

| Initial state | Initial pD | D reaction | Q reaction | Primary superoxide | Escape | Survival | Balance |
|---|---|---|---|---|---|---|---|
| unpolarized I6/6 | 0.3333 333 333 333 3337 | 0.1848 290 544 320 762 | 0.0572 998 597 475 4856 | 0.2421 289 141 796 2476 | 0.757827 651 907 562 | 4.343391 281 261 024e-05 | 0.9999 999 999 999 993 |
| doublet-manifold mixture PD/2 | 0.9999 999 999 999 999 | 0.4333 861 334 320 349 | 0.0121 101 029 864 19 376 | 0.4454 962 364 184 543 | 0.5544 871 632 962 286 | 1.6600 285 316 311 623e-05 | 0.9999 999 999 999 993 |
| quartet-manifold mixture PQ/4 | −1.008102 552 748 629e-17 | 0.0605 505 149 320 96 816 | 0.0798 947 381 281 1315 | 0.1404 452 530 602 0997 | 0.8594 978 962 132 286 | 5.685072 656 075 954e-05 | 0.9999 999 999 999 993 |
| manifold mixture pD=0.25 | 0.2499 999 999 999 9994 | 0.1537 594 195 570 8133 | 0.0629 485 793 426 8971 | 0.2167 079 988 997 7105 | 0.7832 452 129 839 784 | 4.678811 624 964 756e-05 | 0.9999 999 999 999 992 |
| manifold mixture pD=0.75 | 0.7499 999 999 999 999 | 0.3401 772 288 070 5037 | 0.0290 562 617 718 42 826 | 0.3692 334 905 788 932 | 0.6307 398 465 254 785 | 2.6662 895 627 423 608e-05 | 0.9999 999 999 999 992 |

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table04_baseline_results.csv

## Table 5 Controls and Extrema

| Type | Source | Control | Initial state | Mixing/kref | Relaxation/kref | Escape/kref | kQ/kD | Primary superoxide | Escape yield | Survival |
|---|---|---|---|---|---|---|---|---|---|---|
| control | figure07_controls | Baseline sensitivity | | | | | | 0.2421 289 141 7 962 476 | 0.757827 651 907 562 | 4.343391 281 261 024e-05 |
| control | figure07_controls | Zero mixing | | | | | | 0.2338 612 039 677 051 | 0.7660 670 883 725 904 | 7.170765 970 365 598e-05 |
| control | figure07_controls | Spin-independent reaction | | | | | | 0.4999 999 437 324 118 | 0.4999 999 437 3 241 186 | 1.1253 517 471 925 926e-07 |
| control | figure07_controls | Fast relaxation | | | | | | 0.285254 760 496 954 | 0.714731 466 709 809 | 1.3772 793 160 686 254e-05 |
| control | figure07_controls | Rapid escape | | | | | | 0.0039 663 670 09 909 883 | 0.9960 336 329 900 332 | 0.0 |
| control | figure07_controls | Doublet benchmark | | | | | | 0.4454 962 364 184 543 | 0.5544 871 632 962 286 | 1.6600 285 316 311 623e-05 |
| control | figure07_controls | Quartet benchmark | | | | | | 0.1404 452 530 6 020 997 | 0.8594 978 962 132 286 | 5.685072 656 075 954e-05 |
| selected_grid_minimum | figure04_mixing_escape_heatmaps | | unpolarized I6/6 | 0.0 | 0.1 | 100.0 | 0.0 | 0.0033 003 735 153 258 | 0.9966 996 264 846 175 | 0.0 |
| selected_grid_maximum | figure04_mixing_escape_heatmaps | | unpolarized I6/6 | 15.848931 924 611 142 | 0.1 | 0.0 | 10.0 | 0.9999 999 999 999 905 | 0.0 | 2.513571 360 957 245e-15 |

| Type | Source | Control | Initial state | Mixing/kref | Relaxation/kref | Escape/kref | kQ/kD | Primary superoxide | Escape yield | Survival |
|---|---|---|---|---|---|---|---|---|---|---|
| selected_grid_minimum | figure05_mixing_relaxation_heatmaps | | unpolarized I6/6 | 0.0 | 0.0 | 1.0 | 0.0 | 0.1666 666 479 108 041 | 0.8331 096 728 255 349 | 0.0002 236 792 63 659 914 |
| selected_grid_maximum | figure05_mixing_relaxation_heatmaps | | unpolarized I6/6 | 100.0 | 100.0 | 1.0 | 10.0 | 0.8737 914 889 921 969 | 0.1262 085 110 076 903 | 3.014601 656 113 909 e-28 |
| selected_grid_minimum | figure06_reaction_selectivity | | quartet-manifold mixture PQ/4 | 0.0 | 0.1 | 10.0 | 0.0 | 0.0005 948 839 976 204 568 | 0.9994 051 160 023 725 | 1.1921 979 225 581 182e-35 |
| selected_grid_maximum | figure06_reaction_selectivity | | quartet-manifold mixture PQ/4 | 0.0 | 0.1 | 0.1 | 10.0 | 0.9896 245 364 115 824 | 0.0103 750 676 2 991 176 | 3.959585 024 229 424 e-07 |

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table05_controls_and_extrema.csv

## Table 6 Independent Solver Validation

| Observable | Worst absolute error | Worst relative error | Tolerance | Pass | Reference solver | Independent solver | Scope |
|---|---|---|---|---|---|---|---|
| density_matrix | 5.07316 411 303 867e-13 | 1.8731 682 878 912 015e-12 | 2e-09 | True | constant-generator matrix exponential Pade(13) | independent adaptive Dormand-Prince RK5(4) | numerical implementation agreement, not physical validation |
| doublet_population | 1.044438 840 969 164e-12 | 1.044438 840 969 164e-12 | 3e-09 | True | constant-generator matrix exponential Pade(13) | independent adaptive Dormand-Prince RK5(4) | numerical implementation agreement, not physical validation |
| quartet_population | 9.652834 087 603 424e-13 | 9.652834 087 603 424e-13 | 3e-09 | True | constant-generator matrix exponential Pade(13) | independent adaptive Dormand-Prince RK5(4) | numerical implementation agreement, not physical validation |
| survival | 4.69416 172 599 324e-13 | 4.694161 725 993 241e-13 | 3e-09 | True | constant-generator matrix exponential Pade(13) | independent adaptive Dormand-Prince RK5(4) | numerical implementation agreement, not physical validation |
| doublet_reaction_yield | 4.1511 238 890 734 603e-13 | 4.1511 238 890 734 603e-13 | 3e-09 | True | constant-generator matrix exponential Pade(13) | independent adaptive Dormand-Prince RK5(4) | numerical implementation agreement, not physical validation |
| quartet_reaction_yield | 6.47953 912 746 857e-14 | 6.47953 912 746 857e-14 | 3e-09 | True | constant-generator matrix exponential Pade(13) | independent adaptive Dormand-Prince RK5(4) | numerical implementation agreement, not physical validation |
| primary_superoxide_yield | 3.536615 444 943 436e-13 | 3.536615 444 943 436e-13 | 3e-09 | True | constant-generator matrix exponential Pade(13) | independent adaptive Dormand-Prince RK5(4) | numerical implementation agreement, not physical validation |
| escape_yield | 1.5892 842 597 509 116e-13 | 1.5892 842 597 509 116e-13 | 3e-09 | True | constant-generator matrix exponential Pade(13) | independent adaptive Dormand-Prince RK5(4) | numerical implementation agreement, not physical validation |
| reference_probability_accounting | 2.3092 638 912 203 256e-14 | 2.3092 638 912 203 256e-14 | 1e-08 | True | constant-generator matrix exponential Pade(13) | independent adaptive Dormand-Prince RK5(4) | numerical implementation agreement, not physical validation |
| independent_probability_ac | 7.771561 172 376 096e-16 | 7.771561 172 376 096e-16 | 1e-08 | True | constant-generator matrix | independent adaptive | numerical implementation |

| Observable | Worst absolute error | Worst relative error | Tolerance | Pass | Reference solver | Independent solver | Scope |
|---|---|---|---|---|---|---|---|
| counting | | | | | exponential Pade(13) | Dormand-Prince RK5(4) | agreement, not physical validation |

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table06_independent_solver_validation.csv

# Table 7 Circuit Validation

## Table 7A Numerical metrics

| Metric | Layer | Value | Tolerance | Pass |
|---|---|---|---|---|
| numpy_embedding_statevector_error | NumPy embedding | 5.551115123125783e-17 | 1e-12 | True |
| numpy_embedding_doublet_observable_error | NumPy embedding | 0.0 | 1e-12 | True |
| numpy_embedding_quartet_observable_error | NumPy embedding | 0.0 | 1e-12 | True |
| numpy_embedding_leakage_probability | NumPy embedding | 0.0 | 1e-24 | True |
| qiskit_statevector_error | Qiskit | 5.551115123125783e-17 | 1e-12 | True |
| qiskit_doublet_observable_error | Qiskit | 0.0 | 1e-12 | True |
| qiskit_quartet_observable_error | Qiskit | 0.0 | 1e-12 | True |
| qiskit_leakage_probability | Qiskit | 0.0 | 1e-24 | True |
| qiskit_basis_ordering_error | Qiskit | 0.0 | 1e-15 | True |

## Table 7B Execution status and scope

| Metric | Qiskit available | Execution | Implementation | Scope | Reason |
|---|---|---|---|---|---|
| numpy_embedding_statevector_error | True | executed | one inserted dense 8x8 UnitaryGate; not decomposed gates | closed coherent simulator and encoding consistency only | |
| numpy_embedding_doublet_observable_error | True | executed | one inserted dense 8x8 UnitaryGate; not decomposed gates | closed coherent simulator and encoding consistency only | |
| numpy_embedding_quartet_observable_error | True | executed | one inserted dense 8x8 UnitaryGate; not decomposed gates | closed coherent simulator and encoding consistency only | |
| numpy_embedding_leakage_probability | True | executed | one inserted dense 8x8 UnitaryGate; not decomposed gates | closed coherent simulator and encoding consistency only | |
| qiskit_statevector_error | True | executed | one inserted dense 8x8 UnitaryGate; not decomposed gates | closed coherent simulator and encoding consistency only | |

| Metric | Qiskit available | Execution | Implementation | Scope | Reason |
|---|---|---|---|---|---|
| qiskit_doublet_observable_error | True | executed | one inserted dense 8x8 UnitaryGate; not decomposed gates | closed coherent simulator and encoding consistency only | |
| qiskit_quartet_observable_error | True | executed | one inserted dense 8x8 UnitaryGate; not decomposed gates | closed coherent simulator and encoding consistency only | |
| qiskit_leakage_probability | True | executed | one inserted dense 8x8 UnitaryGate; not decomposed gates | closed coherent simulator and encoding consistency only | |
| qiskit_basis_ordering_error | True | executed | one inserted dense 8x8 UnitaryGate; not decomposed gates | closed coherent simulator and encoding consistency only | |

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table07_circuit_validation.csv

# Table 2 Parameter Provenance

The table is split into two aligned panels for print legibility. The symbol column links the panels; the generated CSV preserves the complete row schema.

## Table 2A Identity conditions and value

| Symbol | Definition | Value or range | Unit | Exact species | Charge or protonation | Environment | Temperature | pH |
|---|---|---|---|---|---|---|---|---|
| g_SQ | isotropic doxorubicin semiquinone g value | 2.0035 | dimensionless | doxorubicin semiquinone | protonation/formal charge not assigned | N2-purged 100 mM phosphate; 100 uM doxorubicin; 400 uM xanthine; 0.2 U xanthine oxidase in 2 mL | not reported | 7.5 |
| deltaHpp_SQ | peak-to-peak aqueous one-line width | 3.25 | G | doxorubicin semiquinone | protonation/formal charge not assigned | N2-purged 100 mM phosphate; 100 uM doxorubicin; 400 uM xanthine; 0.2 U xanthine oxidase in 2 mL | not reported | 7.5 |
| deltaHpp_SQ_replication | independent aqueous one-line width | 3.2 | G | doxorubicin semiquinone | protonation/formal charge not assigned | hypoxic 50 mM Tris; 1 mM doxorubicin; 400 uM hypoxanthine; 0.1 U/mL xanthine oxidase | 310 K | 7.4 |
| A_SQ | resolved doxorubicin/adriamycin semiquinone hyperfine values | numeric extraction unavailable | not available | adriamycin semiquinone | not reported in accessible abstract | not reported in accessible abstract | not reported in accessible abstract | not reported in accessible abstract |
| lambda_0 | gas-phase O2 spin-spin fine-structure constant | 1.9847530 | cm^-1 | 16O2 X3Sigma_g- v=0 | neutral | gas phase | not extracted | pH not applicable |
| gamma_0 | gas-phase O2 spin-rotation constant | −8.42930e-3 | cm^-1 | 16O2 X3Sigma_g- v=0 | neutral | gas phase | not extracted | pH not applicable |
| D_O2_gas | axial D from D=2 lambda convention conversion | 3.969506 | cm^-1 | 16O2 X3Sigma_g- v=0 | neutral | gas phase | not extracted | pH not applicable |
| D_O2_solid_air | axial matrix O2 zero-field splitting | 3.572(3) | cm^-1 | ground-state O2 | neutral | solid air condensed in waveguide | 5 K | pH not applicable |
| D_O2_N2_matrix | O2 zero-field splitting in | 3.497 | cm^-1 | ground-state O2 | neutral | O2 diluted in solid N2 | 15 K | pH not applicable |

| Symbol | Definition | Value or range | Unit | Exact species | Charge or protonation | Environment | Temperature | pH |
|---|---|---|---|---|---|---|---|---|
| | solid nitrogen | | | | | | | |
| T1_O2_solution | bulk dissolved oxygen electron spin-lattice relaxation | approximately 7.5 | ps | ground-state O2 | neutral | water; 1:1 water/glycerol; cyclohexane; methanol; benzene; acetone; acetonitrile; DMSO | room temperature | not reported/not applicable |
| K_O2_T4L | O2 association constant in T4 lysozyme L99A cavities | 48 | M^-1 | ground-state O2 | neutral | T4 lysozyme L99A; 50 mM phosphate; 25 mM NaCl | 298 K | 5.5 |
| tau_rot_O2_T4L | O2 rotational correlation components | 0.164;1.41 | ps | ground-state O2 | neutral | T4 lysozyme L99A cavities; 50 mM phosphate; 25 mM NaCl | 298 K | 5.5 |
| g_D_E_O2_encounter | O2 g and ZFS tensors in transient doxorubicin-semiquinone/O2 encounter | unavailable | not available | ground-state O2 | neutral | doxorubicin-semiquinone solution encounter | unavailable | unavailable |
| T1_T2_Tm_SQ | doxorubicin or close anthracycline semiquinone electron-spin relaxation/dephasing | unavailable | s | doxorubicin semiquinone | unresolved | chemically relevant solution | unavailable | unavailable |
| k_O2_total_pH6 | total bulk semiquinone oxidation by O2 | 3.5e8 | M^-1 s^-1 | adriamycin semiquinone plus ground-state O2 | biological-pH semiquinone described as internally protonated/zwitterionic; O2 neutral | aqueous phosphate/formate system; O2 or O2/N2 purge at least 30 min | room temperature | 6.0 |
| k_O2_total_pH11_5 | total bulk semiquinone oxidation by O2 | 1.7e8 | M^-1 s^-1 | adriamycin semiquinone plus ground-state O2 | deprotonated semiquinone form; O2 neutral | aqueous borate/OH- system; O2 or O2/N2 purge at least 30 min | room temperature | 11.5 |
| k_O2_total_1981 | earlier total bulk semiquinone oxidation by O2 | 4.4e7 | M^-1 s^-1 | adriamycin semiquinone plus O2 | not reported in accessible source | aqueous buffer | not reported in accessible source | not reported in accessible source |
| t_half_SQ_anaerobic | semiquinone chemical half-life | 50 | us | adriamycin semiquinone | not reported in accessible source | anaerobic aqueous medium | not reported in accessible source | not reported in accessible source |
| diffusion_distance_SQ | author-calculated bulk travel bounds | <0.6 anaerobic;<0.1 air-saturated | um | adriamycin semiquinone | not reported in accessible source | aqueous buffer | not reported in accessible source | not reported in accessible source |
| encounter_geometry_J_D_kassoc_ksep | separation; orientation; exchange; dipolar; association/separation; cage duration | unavailable | not available | doxorubicin-semiquinone/O2 encounter | unresolved | chemically relevant solution | unavailable | unavailable |
| DQ_coherence | direct spin-correlated coherent D/Q encounter evidence | unavailable | not available | doxorubicin-semiquinone/O2 encounter | unresolved | chemically relevant solution | unavailable | unavailable |
| k_D | doublet encounter electron-transfer rate | unavailable | s^-1 | doublet doxorubicin-semiquinone/O2 encounter | unresolved | chemically relevant solution | unavailable | unavailable |
| k_Q | quartet encounter electron-transfer rate | unavailable | s^-1 | quartet doxorubicin-semiquinone/O2 encounter | unresolved | chemically relevant solution | unavailable | unavailable |
| k_Q_over_k_D | quartet/doublet rate ratio | unavailable | dimensionless | doxorubicin-semiquinone/O2 encounter | unresolved | chemically relevant solution | unavailable | unavailable |
| pKa_HO2 | hydroperoxyl acid dissociation constant | 4.88 | dimensionless | HO2 radical to O2 radical anion | neutral to −1 | oxygen-saturated formic acid/sodium formate aqueous buffer | not reported | 0-13 investigated |
| k_HH | elementary HO2 plus HO2 dismutation rate | 0.76e6 | M^-1 s^-1 | HO2 radical plus HO2 radical | neutral plus neutral | oxygen-saturated formic acid/sodium formate aqueous buffer | not reported | 0-13 mechanism study |
| k_HA | elementary HO2 plus O2 radical-anion dismutation | 8.5e7 | M^-1 s^-1 | HO2 radical plus superoxide | neutral plus −1 | oxygen-saturated formic acid/sodium formate aqueous | not reported | 0-13 mechanism study |

| Symbol | Definition | Value or range | Unit | Exact species | Charge or protonation | Environment | Temperature | pH |
|---|---|---|---|---|---|---|---|---|
| | rate | | | | | buffer | | |
| k_AA | direct superoxide plus superoxide disproportionation upper bound | <100 | M^-1 s^-1 | superoxide plus superoxide | −1 plus −1 | oxygen-saturated formic acid/sodium formate aqueous buffer | not reported | 0-13 mechanism study |
| k_SOD_oxidation | bovine CuZnSOD oxidized-enzyme half-reaction | 1.2e9 | M^-1 s^-1 | bovine CuZnSOD plus superoxide | enzyme state E0 plus −1 | aqueous O2; EDTA; formate | not reported | not reported in accessible record |
| k_SOD_reduction | bovine CuZnSOD reduced-enzyme half-reaction | 2.2e9 | M^-1 s^-1 | bovine CuZnSOD plus superoxide | enzyme state E- plus −1 | aqueous O2; EDTA; formate | not reported | not reported in accessible record |
| k_SOD_overall | overall bovine CuZnSOD-superoxide rate | 2.37e9 | M^-1 s^-1 | native bovine CuZnSOD plus superoxide | enzyme plus −1 | aqueous | 298 K | not reported in accessible abstract |
| v_O2minus_heart_SMP | superoxide formation rate | control 1.6+/-0.2;doxorubicin 69.6+/-2.7 | nmol min^-1 mg^-1 | bovine-heart submitochondrial particles plus doxorubicin | doxorubicin HCl microspecies not reported | isolated submitochondrial particles | not reported in accessible abstract | not reported in accessible abstract |
| v_H2O2_heart_SMP | hydrogen peroxide formation rate | control undetectable;doxorubicin 2.2+/-0.3 | nmol min^-1 mg^-1 | bovine-heart submitochondrial particles plus doxorubicin | doxorubicin HCl microspecies not reported | isolated submitochondrial particles | not reported in accessible abstract | not reported in accessible abstract |
| v_O2minus_Ehrlich_microsomes | SOD-inhibitable superoxide proxy rate; control vs drug | 0.51+/-0.26 n=3;14.71+/-1.43 n=6 | nmol min^-1 mg^-1 | Ehrlich-Lettre tumor microsomes plus doxorubicin | doxorubicin HCl microspecies not reported | 200 ug protein in 1 mL; 150 mM potassium phosphate; 100 uM EDTA; 1 mM NADPH | 310 K | 7.4 |
| v_O2minus_Ehrlich_mitochondria | SOD-inhibitable superoxide proxy rate; control vs drug | 1.12+/-0.15 n=7;7.29+/-0.61 n=10 | nmol min^-1 mg^-1 | Ehrlich tumor mitochondria plus doxorubicin | doxorubicin HCl microspecies not reported | 100 ug protein/mL; sucrose/20 mM HEPES/100 uM EDTA; 100 uM NADH; 4 uM rotenone; 5 min preincubation | 310 K | 8.2 |
| v_O2minus_Ehrlich_nuclei | SOD-inhibitable superoxide proxy rate; control vs drug | 0.36+/-0.05 n=7;3.29+/-0.33 n=12 | nmol min^-1 mg^-1 | Ehrlich tumor nuclear fraction plus doxorubicin | doxorubicin HCl microspecies not reported | 200 ug nuclear protein; sucrose/20 mM HEPES/100 uM EDTA; 1 mM NADPH | 310 K | 7.4 |
| v_H2O2_Ehrlich | H2O2 formation in microsomes; mitochondria; intact cells | 3.42+/-0.48 n=4;2.6+/-0.6 n=3;0.64+/-0.04 n=3 | nmol min^-1 mg^-1; nmol min^-1 per 1e7 cells | Ehrlich tumor fractions or intact cells plus doxorubicin | doxorubicin HCl microspecies not reported | fraction buffers as above; intact 1.5e7 cells/3 mL Chelex-treated PBS | 310 K | 7.4 microsomes;8.2 mitochondria;cell PBS pH not reported |
| H2O2_PC3 | apparent intracellular steady-state hydrogen peroxide | basal 13+/-4;1 uM doxorubicin 51+/-13 | pM | PC3 human prostate cancer cells plus doxorubicin | doxorubicin microspecies not reported | cell culture | 310 K | not reported for intracellular measurement |
| MitoSOX_H9c2 | relative mitochondrial oxidant fluorescence | 1.40+/-0.04;1.80+/-0.08;2.80+/−0.08 | fold control | rat H9c2 cardiomyoblasts plus doxorubicin | doxorubicin microspecies not reported | HBSS with Ca/Mg and 1% BSA; 5 uM MitoSOX | 310 K | not reported |
| negative_5_imino | quinone-disabled anthracycline negative control | doxorubicin 14.71+/-1.43;5-iminodaunorubicin 0.45+/-0.15 | nmol min^-1 mg^-1 | Ehrlich microsomes plus anthracycline | dissolved microspecies not reported | microsomal system above | 310 K | 7.4 |

# Table 2B Method provenance and suitability

| Symbol | Method | DOI or direct source | Status | Uncertainty | Limitations | Suitability | Code location |
|---|---|---|---|---|---|---|---|
| g_SQ | CW EPR; 1 G modulation; 40 G scan; 1 mW | https://doi.org/10.1016/0003-9861(91)90023-C | measured | not reported | isotropic one-line signal; no g tensor | sensitivity_and_initialization | active_model.encounter.g_radical |
| deltaHpp_SQ | CW EPR | https://doi.org/10.1016/0003-9861(91)90023-C | measured | not reported | unresolved hyperfine and homogeneous/inhomogeneous | validation_only | reference_evidence.hyperfine.aqueous_linewidth_pH7_5 |

| Symbol | Method | DOI or direct source | Status | Uncertainty | Limitations | Suitability | Code location |
|---|---|---|---|---|---|---|---|
| | | | | | broadening; cannot uniquely convert to T2 | | |
| deltaHpp_SQ_replication | CW EPR | https://doi.org/10.1124/mol.115.098798 | measured | not reported | unresolved line; not a hyperfine tensor or T2 measurement | validation_only | reference_evidence.hyperfine.aqueous_linewidth_replication |
| A_SQ | EPR; ENDOR; TRIPLE resonance; partial deuteration | https://doi.org/10.1002/mrc.1260260812 | measured_source_identified | not reported in accessible abstract | resolved assignments were investigated; numeric table/tensors/signs/isotope mapping and conditions were not accessible; do not say not investigated | unavailable_for_parameterization | reference_evidence.hyperfine.exact_study |
| lambda_0 | microwave/optical spectral fit | https://hitran.org/media/refs/HITRAN-1973.pdf | fitted | not available in accessible table | gas rovibronic Hamiltonian and convention; never transfer into solution encounter | context_only | reference_evidence.oxygen.gas.lambda_0 |
| gamma_0 | microwave/optical spectral fit | https://hitran.org/media/refs/HITRAN-1973.pdf | fitted | not available in accessible table | gas rovibronic Hamiltonian; never transfer into solution encounter | context_only | reference_evidence.oxygen.gas.gamma_0 |
| D_O2_gas | derived convention conversion | https://hitran.org/media/refs/HITRAN-1973.pdf | calculated | not propagated | not a new measurement; sign and normalization depend on Hamiltonian convention | context_only | reference_evidence.oxygen.gas.derived_D |
| D_O2_solid_air | 94-550 GHz multifrequency EPR | https://doi.org/10.1006/jmre.2000.2175 | measured_fitted | 0.003 cm^-1 | frozen solid only; prohibited as solution surrogate | context_only | reference_evidence.oxygen.matrix_solid_air |
| D_O2_N2_matrix | high-field high-frequency EPR | https://doi.org/10.1016/S0921-4526(00)00615-3 | measured_fitted | not reported | temperature dependent; signal disappears above 25 K; g=2/no rhombicity fit | context_only | reference_evidence.oxygen.matrix_nitrogen |
| T1_O2_solution | proton magnetic relaxation dispersion | https://doi.org/10.1006/jmre.2000.2219 | indirectly_fitted | not reported in abstract | bulk fluid scale; no direct time-domain O2 T1; not encounter-specific and not a static g/D/E tensor | sensitivity_context | reference_evidence.oxygen.bulk_solution |
| K_O2_T4L | gas-pressure NMR plus MD | https://doi.org/10.1038/srep20534 | measured_fitted | 7 M^-1 | protein-cavity association; not doxorubicin/O2 and not a spin-Hamiltonian value | context_only | reference_evidence.oxygen.protein_cavity.association |
| tau_rot_O2_T4L | molecular dynamics fit | https://doi.org/10.1038/srep20534 | calculated_fitted | 0.006;0.02 ps | rapid orientation averaging; not doxorubicin encounter dynamics | context_only | reference_evidence.oxygen.protein_cavity.rotational_correlation |
| g_D_E_O2_encounter | targeted primary-literature search | NA | unavailable | NA | not found; gas/matrix/liquid/protein values are not substitutes | sensitivity_only | reference_evidence.oxygen.doxorubicin_semiquinone_encounter |
| T1_T2_Tm_SQ | pulse-EPR and linewidth search | NA | unavailable | NA | only unresolved CW linewidths found; chemical half-life is not spin relaxation | sensitivity_only | reference_evidence.hyperfine.spin_relaxation |
| k_O2_total_pH6 | electron pulse radiolysis transient absorption | https://doi.org/10.1038/bjc.1985.74 | measured | 0.4e8 M^-1 s^-1 | total channel only; homogeneous solution; no encounter geometry or spin resolution | quantitative_bulk_validation | parameter_records.doxorubicin_semiquinone_plus_oxygen_pH6 |
| k_O2_total_pH11_5 | electron pulse radiolysis transient absorption | https://doi.org/10.1038/bjc.1985.74 | measured | 0.2e8 M^-1 s^-1 | total channel only; alkaline; no encounter geometry or spin resolution | quantitative_bulk_validation | parameter_records.doxorubicin_semiquinone_plus_oxygen_pH11_5 |
| k_O2_total_1981 | pulse radiolysis | https://doi.org/10.1016/0003-9861(81)90263-0 | measured | not reported | differs from 1985 values; missing conditions prevent reconciliation; never k_D | validation_only | parameter_records.doxorubicin_semiquinone_plus_oxygen_1981 |
| t_half_SQ_anaerobic | pulse radiolysis | https://doi.org/10.1016/0003-9861(81)90263-0 | measured | not reported | chemical loss not spin T1/T2 or encounter duration | context_only | reference_evidence.encounter.chemical_lifetime_context |
| diffusion_distance_SQ | calculated from chemical lifetime/diffusion | https://doi.org/10.1016/0003-9861(81)90263-0 | calculated | not reported | not pair separation or encounter radius; 8 us air-saturated | context_only | reference_evidence.encounter.diffusion_distance_context |

| Symbol | Method | DOI or direct source | Status | Uncertainty | Limitations | Suitability | Code location |
|---|---|---|---|---|---|---|---|
| | | | | | chemical lifetime is not cage time | | |
| encounter_geometry_J_D_kassoc_ksep | targeted structure/EPR/kinetics search | NA | unavailable | NA | not reported in inspected exact-system papers and not found elsewhere | sensitivity_only | reference_evidence.encounter.geometry_exchange_dipolar_association_separation_duration |
| DQ_coherence | targeted transient-EPR; CIDEP/CIDNP; magnetic-field; product-branching search | NA | unavailable | NA | retrieved exact-system studies did not investigate D/Q observables; no separate direct paper found | sensitivity_only | reference_evidence.encounter.coherence |
| k_D | spin-resolved kinetics search | NA | unavailable | NA | bulk M^-1 s^-1 total rates cannot be converted or decomposed | illustrative_only | reference_evidence.encounter.k_D |
| k_Q | spin-resolved kinetics search | NA | unavailable | NA | no state-resolved rate or channel yield | illustrative_only | reference_evidence.encounter.k_Q |
| k_Q_over_k_D | spin-resolved kinetics search | NA | unavailable | NA | no defensible physical range or ordering; 0/0.01/0.1/1 are inactive illustrative benchmarks only | illustrative_only | reference_evidence.encounter.k_Q_over_k_D |
| pKa_HO2 | pulse radiolysis | https://doi.org/10.1021/j100711a009 | measured | 0.10 | buffer absence explains disagreement with earlier values; not a membrane/protein pKa | quantitative_aqueous | reference_evidence.downstream.HO2_pKa |
| k_HH | pulse-radiolysis decay | https://doi.org/10.1021/j100711a009 | measured | not reported | species-resolved; do not call an O2- plus O2- rate | quantitative_aqueous | reference_evidence.downstream.k_HO2_HO2 |
| k_HA | pulse-radiolysis decay | https://doi.org/10.1021/j100711a009 | measured | not reported | requires explicit acid-base speciation | quantitative_aqueous | reference_evidence.downstream.k_HO2_O2minus |
| k_AA | pulse-radiolysis bound | https://doi.org/10.1021/j100711a009 | measured_upper_bound | not reported | 2026 reevaluation gives tighter <0.3 M^-1 s^-1; common neutral-pH effective rates are not this elementary step | quantitative_aqueous | reference_evidence.downstream.k_O2minus_O2minus |
| k_SOD_oxidation | pulse radiolysis optical 650/300 nm | https://doi.org/10.1021/ja00790a007 | measured | 0.2e9 M^-1 s^-1 | exact enzyme/preparation only; surrogate for human or mitochondrial SOD | sensitivity_benchmark | reference_evidence.downstream.k_CuZnSOD_oxidation |
| k_SOD_reduction | pulse radiolysis optical 650/300 nm | https://doi.org/10.1021/ja00790a007 | measured | 0.4e9 M^-1 s^-1 | exact enzyme/preparation only; surrogate otherwise | sensitivity_benchmark | reference_evidence.downstream.k_CuZnSOD_reduction |
| k_SOD_overall | pulse radiolysis plus optical/EPR | https://doi.org/10.1042/bj1390049 | measured | 0.18e9 M^-1 s^-1 | not a universal intracellular SOD1/SOD2 constant | sensitivity_benchmark | reference_evidence.downstream.k_CuZnSOD_overall |
| v_O2minus_heart_SMP | SOD-inhibitable cytochrome-c reduction | https://doi.org/10.1016/S0021-9258(17)35747-2 | measured_mean_SE | 0.2;2.7 | 90 uM dose; exact buffer oxygen and timing unavailable in accessible abstract | quantitative_matched_validation | validation_datasets.beef_heart_submitochondrial_superoxide |
| v_H2O2_heart_SMP | catalase-released oxygen | https://doi.org/10.1016/S0021-9258(17)35747-2 | measured_mean_SE | 0.3 | 200 uM dose; detection limit and exact conditions not reported in accessible abstract | quantitative_matched_validation | validation_datasets.beef_heart_submitochondrial_H2O2 |
| v_O2minus_Ehrlich_microsomes | acetylated cytochrome-c reduction at 550 nm | https://doi.org/10.1155/2019/9474823 | measured_mean_SE | 0.26;1.43 | 135 uM dose; initial linear rate; oxygenation not separately reported | quantitative_matched_validation | validation_datasets.Ehrlich_microsome_superoxide |
| v_O2minus_Ehrlich_mitochondria | acetylated cytochrome-c reduction at 550 nm | https://doi.org/10.1155/2019/9474823 | measured_mean_SE | 0.15;0.61 | 135 uM dose; alkaline and rotenone-blocked; oxygenation not separately reported | quantitative_matched_validation | validation_datasets.Ehrlich_mitochondrial_superoxide |
| v_O2minus_Ehrlich_nuclei | acetylated cytochrome-c reduction at 550 nm | https://doi.org/10.1155/2019/9474823 | measured_mean_SE | 0.05;0.33 | 135 uM dose; isolated fraction; oxygenation not separately reported | quantitative_matched_validation | validation_datasets.Ehrlich_nuclear_superoxide |
| v_H2O2_Ehrlich | catalase-released oxygen; linear 10-30 min interval | https://doi.org/10.1155/2019/9474823 | measured_mean_SE | 0.48;0.6;0.04 | 135 uM fractions;400 uM cells; controls undetectable with detection limits not reported; | quantitative_matched_validation | validation_datasets.Ehrlich_H2O2 |

| Symbol | Method | DOI or direct source | Status | Uncertainty | Limitations | Suitability | Code location |
|---|---|---|---|---|---|---|---|
| | | | | | PBS air-bubbled 30 min | | |
| H2O2_PC3 | catalase-aminotriazole kinetic assay | https://doi.org/10.1016/j.abb.2005.06.015 | measured_derived | 4;13 pM | 30 min exposure; oxygen fraction not reported; extracellular Amplex Red did not increase; catalase overexpression did not improve resistance | quantitative_matched_validation | validation_datasets.PC3_intracellular_H2O2 |
| MitoSOX_H9c2 | 1 h fluorescence | https://doi.org/10.1016/j.bbrc.2007.04.106 | measured_mean_SD | 0.04;0.08;0.08 | 10/20/50 uM doses; n=3; oxygen not reported; doxorubicin autofluorescence and nonspecific probe products prohibit absolute calibration | sensitivity_only | validation_datasets.H9c2_MitoSOX |
| negative_5_imino | acetylated cytochrome-c reduction | https://doi.org/10.1155/2019/9474823 | measured_mean_SE | 1.43;0.15 | both 135 uM; supports quinone dependence but not D/Q selectivity | qualitative_disconfirming | validation_datasets.quinone_disabled_negative_control |

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# Table 3 Unavailable Parameters and Consequences

| Unavailable parameter | Why needed | Sensitivity allowed | Range status | Allowed treatment | Prohibited conclusion |
|---|---|---|---|---|---|
| accessible numerical doxorubicin semiquinone hyperfine tensor | Defines nuclear-spin-driven electron-spin mixing in the exact radical. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits an exact-system hyperfine Hamiltonian or mixing rate. |
| encounter k_doublet_s | Sets doublet-resolved electron-transfer flux. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits a quantitative D-channel reaction yield. |
| encounter k_quartet_s | Sets quartet-resolved electron-transfer flux. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits a quantitative Q-channel reaction yield. |
| physical k_quartet_s/k_doublet_s range | Determines reaction selectivity between the two manifolds. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits assigning a preferred manifold or physical selectivity range. |
| encounter escape rate | Sets competition between reaction and encounter separation. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits a quantitative reaction-versus-escape branching ratio. |
| encounter duration | Sets a physical observation endpoint or residence time. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits a physical encounter time axis and endpoint. |
| semiquinone T1 | Defines longitudinal relaxation within the encounter. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits an encounter-specific relaxation rate. |
| semiquinone T2/Tm | Defines coherence decay under a physically matched model. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits quantitative dephasing or coherence-lifetime claims. |
| exchange coupling | Sets the isotropic electron-electron coupling magnitude. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits a quantitative exchange Hamiltonian. |
| dipolar coupling and geometry | Sets anisotropic coupling and requires geometry/orientation. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits a quantitative dipolar Hamiltonian and orientation average. |
| solution encounter O2 g and ZFS tensors | Defines encounter-specific oxygen Zeeman/ZFS evolution. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits transfer of gas, matrix, or protein O2 tensors into solution. |

| Unavailable parameter | Why needed | Sensitivity allowed | Range status | Allowed treatment | Prohibited conclusion |
|---|---|---|---|---|---|
| encounter-specific O2 relaxation | Defines spin relaxation of O2 within the solution encounter. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits assigning a solution-encounter O2 relaxation rate. |
| compartment-specific H2O2 loss rate | Defines compartment-specific removal of downstream H2O2. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits compartment-level H2O2 prediction. |
| encounter association rate | Connects bulk semiquinone and oxygen concentrations to encounter formation. | yes, with explicit opt-in | illustrative only; no defensible physical range | bounded illustrative sensitivity coordinate only | Prohibits a quantitative encounter-formation flux. |
| encounter geometry separated from dipolar coupling | Defines separation, orientation, association, and residence physics. | yes, with explicit opt-in | illustrative only; no defensible physical range | illustrative geometry scenarios only; none used in the baseline | Prohibits a physical association or encounter model. |
| chemically prepared initial D/Q population | Defines the encounter birth density matrix. | yes, as benchmark mixtures only | illustrative only; no measured preparation distribution | unpolarized and manifold-mixture benchmarks only | Prohibits choosing a chemically prepared initial D/Q state. |

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# Table 8 Experimental Validation Evidence

## Table 8A System conditions and reported result

| Record | Exact system | Observable | Conditions | Reported result | Unit |
|---|---|---|---|---|---|
| beef_heart_submitochondrial_superoxide | beef_heart_submitochondrial_superoxide | beef | dose=90 uM doxorubicin; oxygen=not reported in accessible abstract | {"control": 1.6, "doxorubicin": 69.6, "uncertainty_SE": [0.2, 2.7]} | nmol min^-1 mg^-1 |
| beef_heart_submitochondrial_H2O2 | beef_heart_submitochondrial_H2O2 | beef | dose=200 uM doxorubicin; oxygen=not reported in accessible abstract | {"control": "undetectable", "doxorubicin": 2.2, "uncertainty_SE": 0.3} | nmol min^-1 mg^-1 |
| Ehrlich_microsome_superoxide | 200 ug microsomal protein in 1 mL; 150 mM potassium phosphate; 100 uM EDTA; 1 mM NADPH | Ehrlich | dose=135 uM; environment=200 ug microsomal protein in 1 mL; 150 mM potassium phosphate; 100 uM EDTA; 1 mM NADPH; temperature=310 K; pH=7.4; oxygen=not separately reported | {"control": 0.51, "doxorubicin": 14.71, "n": [3, 6], "uncertainty_SE": [0.26, 1.43]} | nmol min^-1 mg^-1 |
| Ehrlich_mitochondrial_superoxide | 100 ug protein/mL; 250 mM sucrose; 20 mM HEPES; 100 uM EDTA; NADH; 4 uM rotenone; 5 min preincubation | Ehrlich | dose=135 uM; environment=100 ug protein/mL; 250 mM sucrose; 20 mM HEPES; 100 uM EDTA; NADH; 4 uM rotenone; 5 min preincubation; temperature=310 K; pH=8.2; oxygen=not separately reported | {"control": 1.12, "doxorubicin": 7.29, "n": [7, 10], "uncertainty_SE": [0.15, 0.61]} | nmol min^-1 mg^-1 |
| Ehrlich_nuclear_superoxide | 200 ug nuclear protein; sucrose/20 mM HEPES/100 uM EDTA; 1 mM NADPH | Ehrlich | dose=135 uM; environment=200 ug nuclear protein; sucrose/20 mM HEPES/100 uM EDTA; 1 mM NADPH; temperature=310 K; pH=7.4; oxygen=not separately reported | {"control": 0.36, "doxorubicin": 3.29, "n": [7, 12], "uncertainty_SE": [0.05, 0.33]} | nmol min^-1 mg^-1 |
| Ehrlich_H2O2 | fraction buffers as separately reported; cells in Chelex-treated PBS | Ehrlich | dose=135 uM fractions; 400 uM cells; environment=fraction buffers as separately reported; cells in Chelex-treated PBS; temperature=310 K; pH=7.4 microsomes;8.2 mitochondria;cell PBS pH not reported; oxygen=cells air-bubbled 30 | {"uncertainty_SE": [0.48, 0.6, 0.04], "values": [3.42, 2.6, 0.64]} | nmol min^-1 mg^-1; nmol min^-1 per 1e7 cells |

| Record | Exact system | Observable | Conditions | Reported result | Unit |
|---|---|---|---|---|---|
| | | | min | | |
| PC3_intracellular_H2O2 | PC3 human prostate cancer cells | PC3 | dose=1 uM doxorubicin; environment=PC3 human prostate cancer cells; temperature=310 K; oxygen=not reported | {"basal": 13, "doxorubicin": 51, "time": "30 min", "uncertainty": [4, 13]} | pM |
| H9c2_MitoSOX | HBSS with Ca/Mg and 1% BSA; 5 uM MitoSOX | H9c2 | dose=10/20/50 uM doxorubicin; environment=HBSS with Ca/Mg and 1% BSA; 5 uM MitoSOX; temperature=310 K; oxygen=not reported | {"uncertainty": [0.04, 0.08, 0.08], "values": [1.4, 1.8, 2.8]} | fold control |
| quinone_disabled_negative_control | Ehrlich microsomal system | quinone | dose=135 uM both compounds; environment=Ehrlich microsomal system; temperature=310 K; pH=7.4; oxygen=not separately reported | {"doxorubicin": 14.71, "five_iminodaunorubicin": 0.45, "uncertainty_SE": [1.43, 0.15]} | nmol min^-1 mg^-1 |

## Table 8B Evidence method and limitation

| Record | Method | Source | Direct validation | Limitation |
|---|---|---|---|---|
| beef_heart_submitochondrial_superoxide | SOD-inhibitable cytochrome-c reduction | doi:10.1016/S0021-9258(17)35747-2 | no | Conditions and normalization differ; do not pool or fit. |
| beef_heart_submitochondrial_H2O2 | catalase-released oxygen | doi:10.1016/S0021-9258(17)35747-2 | no | Detection limit and exact conditions not reported in accessible abstract. |
| Ehrlich_microsome_superoxide | SOD-inhibitable acetylated cytochrome-c reduction | doi:10.1155/2019/9474823 | no | Conditions and normalization differ; do not pool or fit. |
| Ehrlich_mitochondrial_superoxide | SOD-inhibitable acetylated cytochrome-c reduction | doi:10.1155/2019/9474823 | no | Conditions and normalization differ; do not pool or fit. |
| Ehrlich_nuclear_superoxide | SOD-inhibitable acetylated cytochrome-c reduction | doi:10.1155/2019/9474823 | no | Conditions and normalization differ; do not pool or fit. |
| Ehrlich_H2O2 | catalase-released oxygen; linear 10-30 min interval | doi:10.1155/2019/9474823 | no | Conditions and normalization differ; do not pool or fit. |
| PC3_intracellular_H2O2 | catalase-aminotriazole kinetic assay | doi:10.1016/j.abb.2005.06.015 | no | Apparent intracellular H2O2; medium Amplex Red did not increase and catalase overexpression did not increase resistance. |
| H9c2_MitoSOX | 1 h fluorescence | doi:10.1016/j.bbrc.2007.04.106 | no | Sensitivity only; autofluorescence and nonspecific probe products prohibit absolute calibration. |
| quinone_disabled_negative_control | acetylated cytochrome-c reduction | doi:10.1155/2019/9474823 | no | Supports quinone dependence but not D/Q selectivity. |

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# Table 9 Requirements Traceability

## Table 9A Requirement to implementation mapping

| Requirement | Topic or output | Exact file | Exact code object | Figure or table |
|---|---|---|---|---|
| Topic 1 | Semiquinone g and hyperfine | configs/doxorubicin_parameters.json; spin_chemistry.py | hamiltonian; validate_authority_bundle | Figure 7; Table 2 |
| Topic 2 | O2 electronic parameters | configs/doxorubicin_parameters.json | active_model.encounter.g_oxygen; oxygen_zfs_d_rad_s; oxygen_zfs_e_rad_s | Tables 2-3 |
| Topic 3 | Spin relaxation | spin_chemistry.py | _lindblad; propagate_encounter_reference; propagate_encounter_independent | Figures 5, 7, 8 |
| Topic 4 | Bulk SQ/O2 kinetics | spin_chemistry.py; configs/parameter_provenance.csv | evaluate_bulk_superoxide_rate | Figure 10; Table 2 |
| Topic 5 | Encounter coherence | spin_chemistry.py | hamiltonian; propagate_encounter_reference | Figures 1-2 |
| Topic 6 | D/Q reaction selectivity | spin_chemistry.py; sensitivity_analysis.py | P_DOUBLET; P_QUARTET; run_selectivity_sweep | Figures 3, 4, 6 |
| Topic 7 | Escape and encounter lifetime | spin_chemistry.py; sensitivity_analysis.py | EncounterParameters.k_escape_s; run_mixing_escape_sweep | Figures 4, 7; Table 5 |
| Topic 8 | Exchange dipolar and geometry | spin_chemistry.py | EncounterParameters; hamiltonian | Figure 7; Table 3 |
| Topic 9 | Downstream speciation and dismutation | spin_chemistry.py | downstream_ros_species_resolved | Figure 9 |
| Topic 10 | Experimental ROS validation | configs/doxorubicin_parameters.json | validation_datasets | Table 8 |
| Topic 11 | Independent numerical behavior | spin_chemistry.py | propagate_encounter_reference; propagate_encounter_independent | Figure 8; Table 6 |
| Topic 12 | Circuit consistency | spin_chemistry.py | execute_three_qubit_unitary_circuit | Figure 11; Table 7 |
| Output A | parameter provenance | configs/parameter_provenance.csv | _provenance_rows | Table 2 |
| Output B | Hamiltonian and spin algebra | spin_chemistry.py | hamiltonian; P_DOUBLET; P_QUARTET | Figure 1; Table 1 |
| Output C | initial-state benchmarks | spin_chemistry.py | initial_density | Figures 2-3; Table 4 |
| Output D | kQ/kD analysis | sensitivity_analysis.py | run_mixing_escape_sweep; run_selectivity_sweep | Figures 4 and 6 |
| Output E | reaction network | spin_chemistry.py | primary_superoxide_formation; downstream_ros_species_resolved | Figures 1 and 9 |
| Output F | supported claims | deliverables/ROS_Spin_professor_summary.md | SUPPORTED_CLAIMS | Professor summary |
| Output G | unsupported claims | deliverables/ROS_Spin_professor_summary.md | PROHIBITED_CLAIMS | Professor summary |
| Output H | configuration JSON | configs/doxorubicin_parameters.json | validate_authority_bundle | Table 2 |
| Output I | machine-readable outputs | paper_analysis.py | write_csv; run_manifest.json | Figures 1-11; Tables 1-9 |
| Output J | existing repository and legacy mapping | README.md; legacy_entrypoint.py | retired_entrypoint | Table 9 |

# Table 9B Test evidence and remaining limitation

| Requirement | Test | Evidence status | Remaining limitation |
|---|---|---|---|
| Topic 1 | InitialStateAndHamiltonianTests; AuthorityAndModeTests | bounded_complete | isotropic g only; hyperfine tensors unavailable |
| Topic 2 | AuthorityAndModeTests | evidence_review_complete | encounter tensors unavailable |
| Topic 3 | EncounterDynamicsTests; IndependentEncounterValidationTests | framework_complete | T1 and T2 unavailable |
| Topic 4 | AuthorityAndModeTests; PaperPipelineTests | condition_specific_complete | not encounter kD or kQ |
| Topic 5 | EncounterDynamicsTests | evidence_unavailable | no measured coherent encounter |
| Topic 6 | SpinAlgebraTests; PaperPipelineTests | sensitivity_complete | no physical ordering known |
| Topic 7 | EncounterDynamicsTests | sensitivity_complete | physical value unavailable |
| Topic 8 | InitialStateAndHamiltonianTests | framework_complete | magnitudes and geometry unavailable |
| Topic 9 | StagedChemistryTests | bounded_complete | fixed pH and constant SOD |
| Topic 10 | AuthorityAndModeTests | catalog_complete | datasets incomparable for fitting |
| Topic 11 | IndependentEncounterValidationTests | complete | numerical, not physical validation |
| Topic 12 | CoherentValidationTests | conditional_complete | coherent simulator and embedding only |
| Output A | AuthorityAndModeTests | complete | Interpret only within the stated evidence and sensitivity scope. |
| Output B | SpinAlgebraTests; InitialStateAndHamiltonianTests | complete | Interpret only within the stated evidence and sensitivity scope. |
| Output C | InitialStateAndHamiltonianTests | complete | Interpret only within the stated evidence and sensitivity scope. |
| Output D | PaperPipelineTests | complete | Interpret only within the stated evidence and sensitivity scope. |
| Output E | StagedChemistryTests | complete | Interpret only within the stated evidence and sensitivity scope. |
| Output F | test_professor_handoff.py | complete | Interpret only within the stated evidence and sensitivity scope. |
| Output G | test_professor_handoff.py | complete | Interpret only within the stated evidence and sensitivity scope. |
| Output H | AuthorityAndModeTests | complete | Interpret only within the stated evidence and sensitivity scope. |
| Output I | PaperPipelineTests | complete | Interpret only within the stated evidence and sensitivity scope. |
| Output J | PaperPipelineTests | complete | Interpret only within the stated evidence and sensitivity scope. |

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## Supported Claims

- The spin-1/2 semiquinone and spin-1 ground-state oxygen electronic space has dimension six and decomposes into doublet and quartet manifolds of dimensions two and four.
- The implemented projectors, density matrices, Hamiltonian construction, trace-decreasing evolution, and probability accounting satisfy the repository's algebraic and numerical tests.
- Under declared dimensionless scenarios, primary-superoxide yield depends on competition among spin-selective electron transfer, doublet-quartet mixing, relaxation, and encounter escape.
- The adaptive Dormand-Prince implementation agrees with the constant-generator matrix-exponential reference within the declared numerical tolerances.
- The three-qubit calculation checks coherent six-state-to-eight-state encoding and simulator consistency when Qiskit executes.
- Condition-specific literature bulk rates and downstream aqueous kinetics can provide bounded context when their units, species, and experimental conditions remain explicit.

## Unsupported and Prohibited Claims

- Absolute cellular superoxide or absolute in-vivo H2O2 concentration
- Cardiotoxicity or therapeutic response
- A measured kQ/kD value, ordering, probability distribution, or physical range
- A confirmed coherent or spin-correlated semiquinone-oxygen encounter
- A demonstrated doublet, quartet, entangled, or otherwise prepared encounter state
- A quantitative magnetic-field effect
- Transfer of bulk M^-1 s^-1 rates into encounter-level kD or kQ in s^-1
- Quantum advantage, hardware performance, or independent validation of the chemistry

## Recommended Manuscript Framing

Frame the work as an evidence-gated computational framework and conditional sensitivity study connected to the original anthracycline ROS question. Lead with the physical correction, present computed structural validation separately from sensitivity results and literature-derived calculations, and state the missing measurements as requirements for future quantitative chemistry rather than filling them with analogues.

Recommended manuscript language: The simulations demonstrate how primary superoxide yield would depend on the competition among spin-selective electron transfer, doublet-quartet mixing, relaxation, and encounter escape under specified dimensionless scenarios. They do not establish that the assumed spin preparation, mixing magnitude, or state-selective rates occur in the doxorubicin semiquinone-oxygen system.

## Exact Reproduction Instructions

From the repository root, run the following commands after installing the pinned requirements and ensuring LibreOffice is available:

```
.venv/bin/python paper_analysis.py --mode full --output-dir results/paper --execute-circuit --
overwrite
.venv/bin/python render_equations.py assets/equations
python3 professor_handoff.py --results-dir results/paper --deliverables-dir deliverables --
overwrite
```

The exact numerical invocation, dependency versions, source hashes, assumptions, tolerances, Qiskit status, and limitations are recorded in results/paper/metadata/run_manifest.json.

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# Appendix Figure Captions and Table Notes

Each figure caption below is copied from the generated caption file that accompanies the plotted source data.

## Figure 1 Corrected Model Overview

Figure 1. Corrected model overview. Doxorubicin reduction and semiquinone formation are classical upstream stages. A semiquinone-triplet-O2 encounter enters the six-state doublet/quartet quantum spin-evolution stage, followed by competing spin-selective electron transfer or escape. Integrated reaction flux forms primary superoxide; fixed-pH HO2/O2-minus speciation and dismutation subsequently form H2O2, which may be lost through a separate sink. Conceptual workflow created by the authors; not a simulation output. The diagram is non-quantitative and does not assert coherent preparation or a measured encounter.

**Figure source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure01_model_overview.csv

## Figure 2 Benchmark Encounter Trajectories

Figure 2. Benchmark encounter trajectories. Surviving doublet and quartet populations, total survival, cumulative doublet and quartet reaction yields, primary-superoxide yield, and escape yield are shown against normalized time for five benchmark mixtures. At every sampled time, YD + YQ + Yescape + Psurvival = 1 within numerical tolerance. The settings kD/kref=1, kQ/kD=0.1, kescape/kref=1, omega_local/kref=1, and gammaR/kref=gammaO/kref=0.1 are illustrative sensitivity coordinates, not measured parameters.

**Figure source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure02_benchmark_trajectories.csv

## Figure 3 Initial State Comparison

Figure 3. Initial-state comparison. Final reaction, escape, and unresolved survival outcomes are compared for the unpolarized encounter, doublet-manifold benchmark, quartet-manifold benchmark, and pD=0.25 and pD=0.75 mixtures. The doublet and quartet cases are computational limiting benchmarks rather than demonstrated chemically prepared states. Primary superoxide is the sum of integrated D and Q reaction yields, not a final spin population.

**Figure source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure03_initial_state_comparison.csv

## Figure 4 Mixing versus Escape Heatmaps

Figure 4. Mixing-versus-escape heatmaps. Primary-superoxide yield per encounter is calculated over normalized local D/Q mixing and escape rates for kQ/kD=0, 0.1, 1, and 10. The kQ/kD=1 panel is the spin-independent null condition. Zero and logarithmic nonzero coordinates are illustrative; no displayed range is experimentally established.

**Figure source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure04_mixing_escape_heatmaps.csv

## Figure 5 Mixing versus Relaxation Heatmaps

Figure 5. Mixing-versus-relaxation heatmaps. Primary-superoxide yield is calculated for an unpolarized benchmark over normalized local mixing and equal radical/O2 relaxation rates. The fixed settings are kD/kref=1, kescape/kref=1, and duration=8/kref; panels show kQ/kD=0, 0.1, 1, and 10, with equality marked as the spin-independent null. All coordinates are illustrative and non-predictive.

**Figure source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure05_mixing_relaxation_heatmaps.csv

## Figure 6 Reaction Selectivity Sensitivity

Figure 6. Reaction-selectivity sensitivity. Primary-superoxide yield is plotted against kQ/kD for unpolarized, doublet, quartet, pD=0.25, and pD=0.75 benchmarks across three mixing and escape settings. The vertical line marks kQ/kD=1 as the spin-independent null. The kQ/kD axis is a sensitivity coordinate, not a measured or chemically defensible probability distribution or physical range.

**Figure source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure06_reaction_selectivity.csv

## Figure 7 Controls and Limiting Cases

Figure 7. Controls and limiting cases. Baseline sensitivity, zero mixing, kQ=kD spin-independent reaction, fast relaxation, rapid escape, doublet benchmark, and quartet benchmark cases show primary reaction, escape, and unresolved survival. Stacked outcomes make probability accounting visible. All rates and initial-state restrictions are computational controls, not established encounter parameters or preparations.

**Figure source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure07_controls.csv

## Figure 8 Independent Numerical Validation

Figure 8. Independent numerical validation. A separately constructed adaptive Dormand-Prince 5(4) solver is compared with the constant-generator matrix-exponential reference for full density matrices, doublet and quartet populations, survival, reaction yields, escape yield, and probability balance. Dashed lines show declared tolerances. Agreement validates numerical implementation only, not physical encounter inputs.

**Figure source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure08_solver_validation.csv

## Figure 9 Downstream ROS Kinetics

Figure 9. Downstream ROS kinetics. Total superoxide-family radical pool, HO2, O2-minus, accumulated H2O2, and accumulated H2O2 loss are shown for spontaneous dismutation, SOD-mediated dismutation, and SOD plus H2O2 loss. The calculation starts from an illustrative 10 micromolar single radical pulse at fixed pH 7.4, assumes rapid acid-base equilibrium and constant rates/SOD, and consumes two radical equivalents per H2O2. No spin population is mapped directly to H2O2, and the trajectories are not cellular predictions.

**Figure source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure09_downstream_kinetics.csv

## Figure 10 Literature Bulk Rate Comparison

Figure 10. Literature bulk-rate comparison. Actual numerical values stored in the configuration JSON and matched provenance CSV are plotted with reported uncertainty, pH, temperature, environment, species/protonation, method, and source identifiers. Missing conditions remain visibly marked and records are not combined into a universal rate. These M^-1 s^-1 bulk total constants are not encounter-level kD or kQ in s^-1.

**Figure source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure10_bulk_rate_comparison.csv

## Figure 11 Three Qubit Coherent Embedding Validation

Figure 11. Three-qubit coherent-embedding validation. The six-state coherent reference unitary is compared with its eight-state three-qubit embedding and, when requested and available, actual Qiskit statevector execution. Statevector, doublet and quartet observable, basis-ordering, and unused-state leakage errors are compared with declared tolerances. This validates coherent simulator and encoding consistency only; it does not validate reaction, relaxation, escape, downstream chemistry, hardware performance, or quantum advantage.

**Figure source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/data/figure11_circuit_validation.csv

## Table 1 Original versus Corrected ROS Spin Model

Generated programmatically from repository code or machine-readable evidence. Interpret all values within the status, conditions, suitability, and limitations columns.

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table01_original_versus_corrected_model.csv

## Table 2 Parameter Provenance

Generated programmatically from repository code or machine-readable evidence. Interpret all values within the status, conditions, suitability, and limitations columns.

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table02_parameter_provenance.csv

## Table 3 Unavailable Parameters and Consequences

Generated programmatically from repository code or machine-readable evidence. Interpret all values within the status, conditions, suitability, and limitations columns.

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table03_unavailable_parameters_and_consequences.csv

## Table 4 Baseline Results

Generated programmatically from repository code or machine-readable evidence. Interpret all values within the status, conditions, suitability, and limitations columns.

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table04_baseline_results.csv

## Table 5 Controls and Extrema

Generated programmatically from repository code or machine-readable evidence. Interpret all values within the status, conditions, suitability, and limitations columns.

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table05_controls_and_extrema.csv

## Table 6 Independent Solver Validation

Generated programmatically from repository code or machine-readable evidence. Interpret all values within the status, conditions, suitability, and limitations columns.

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table06_independent_solver_validation.csv

## Table 7 Circuit Validation

Generated programmatically from repository code or machine-readable evidence. Interpret all values within the status, conditions, suitability, and limitations columns.

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## Table 8 Experimental Validation Evidence

Generated programmatically from repository code or machine-readable evidence. Interpret all values within the status, conditions, suitability, and limitations columns.

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table08_experimental_validation_evidence.csv

## Table 9 Requirements Traceability

Generated programmatically from repository code or machine-readable evidence. Interpret all values within the status, conditions, suitability, and limitations columns.

**Generated CSV source** /Users/arjunlingala/Downloads/ROS_Spin/results/paper/tables/table09_requirements_traceability.csv